

Lean Proof Strategy Proposal for the Scheduling Rules Document

Stuart Swan, Platform Architect, Siemens EDA

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0. Goal and scope

The goal is to formalize the scheduling rules document's **mathematical proof core** in Lean: trace compatibility, witness construction, cutpoint correspondence, elaboration, and one-way liveness/deadlock preservation. The Lean proof would certify the formalized theorem stack under the stated assumptions, not every informal prose statement in the document.

The scheduling rules document's own framing is conditional: the formal results depend on assumptions, observation-boundary choices, witness-selection conditions, and liveness/progress premises.

The Lean development should therefore be organized around explicit theorem instances and assumption bundles, rather than hidden global axioms.

The primary scheduling rules document on which this doc is based is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf

The latest version of this Lean proof strategy document is available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/Lean_proof_strategy.pdf

The Matchlib examples kit that contains training slides, examples, etc., is available here:

<https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/tree/master>

1. Central comparison object

The proof should be parameterized by a selected comparison instance:

structure ComparisonInstance where

Sys_ref : System

RTL_B : System

-- Shared proof-internal observable alphabet for the Sys_ref / RTL_B comparison.
-- Both sides emit events in this same Σ schema.

Sigma : Alphabet

-- Optional outer DUT/testbench-visible projection alphabet.
-- In ordinary cases this may be the identity projection from Sigma.

Sigma_DUT : Alphabet

sigmaDUTProjection : ProjectionMap Sigma Sigma_DUT

env : ReactiveEnvironment Sigma
refChoice : SysRefChoice
combNetwork? : Option CombNetwork
assumptions : AssumptionBundle
witness : WitnessBundle
contracts : ModelingContracts

The comparison relation between Sys_ref and RTL_B is defined over the single shared proof-internal alphabet C.Sigma. The reference and post-HLS executions may have different internal ε -structure, bucket decomposition, or implementation state, but their observable events are encoded in the same Σ event schema before CompatibleP / CompatibleSys is applied.

Sigma_DUT is not a second post-side alphabet. It is only the chosen outer DUT/testbench-visible projection of C.Sigma. In elaborated cases, introduced proof-internal staged/bundle/join events may be hidden or accounted for by sigmaDUTProjection / Π_{elab} when projecting to Sigma_DUT, while the proof-internal compatibility theorem still ranges over C.Sigma.

Sys_ref is the chosen source reference model for the theorem instance:

- ordinary bounded-FIFO case: Sys_ref = Sys_B;
- elaborated case: Sys_ref = Sys_B^{elab}(E) for a well-formed elaboration package E.

This follows the updated Appendix Q direction: the formal theorems do not require a unique algorithm that constructs Sys_B^{elab}; instead, the chosen Sys_ref is part of the theorem instance.

2. Foundation layer: bucketed executions, not flat traces

The primary execution model should be **cycle-bucketed**, not a flat sequence of $\text{eps} \mid \text{tick} \mid \text{sigma } e$. The reason is that the document allows multiple same-cycle Σ -events without imposing artificial order among independent same-cycle events. The earlier review correctly noted that a flat trace would either impose a fake linearization or require an added bucket abstraction anyway.

Use:

structure State where

cycle : Nat

epsPos : Nat

store : Store

with explicit advancement rules:

- Intra-cycle ε -normalization steps advance epsPos only.
- Same-cycle Σ -events advance epsPos only.
- Clock ticks advance cycle and reset epsPos.
- Cycle-advancing silent progress is represented at the bucket level: a
- CycleBucket with $\text{sigmaEvents} = \emptyset$ may still consume a real clock cycle and
- may represent HLS-inserted FSM latency, arbitration/bookkeeping latency,
- failed non-blocking polls, or pipeline advances that emit no Σ -event.

Define buckets:

structure CycleBucket (Σ : Type) where

cycle : Nat

startState : State

epsPrefix : List EpsStep

sigmaEvents : Finset Σ

sigmaOrder : PartialOrderOn sigmaEvents

epsSuffix : List EpsStep

endState : State

endQuiescent : Prop

def ExecutionBucketed (Σ : Type) :=

List (CycleBucket Σ)

def InfiniteExecutionBucketed (Σ : Type) :=

Nat $\rightarrow$ CycleBucket Σ

This provides the right foundation for:

- ε -quiescent cutpoints;
- rendezvous same-cycle commits;
- R2C combinational fixed-point buckets;
- J.1 ideal induction over unordered or partially ordered same-cycle units;
- B2/B3 temporal reasoning over infinite bucketed executions.

Important separation: ExecutionBucketed is the semantic carrier for state advancement, ε -quiescence, occupancy updates, R2C fixed points, and fairness/progress reasoning. It is not itself the cross-model trace-compatibility predicate.

In particular, system-level compatibility $\tau_{\text{ref},\text{system}} \approx_{\text{sys}} \tau_{\text{post},\text{system}}$ must not be defined as equality of CycleBucket lists, equality of bucket cycle numbers, or bucket-by-bucket equality of sigmaEvents. The document's Appendix J compatibility relation is an ordered cross-model observable-compatibility predicate. It preserves the explicitly required happens-before / $\leq_J$ constraints, atomic multi-endpoint units, E1–E5 obligations, and matched value labels, while allowing independent observable units to commute or to be grouped into different cycle buckets in Sys_ref and RTL_B.

Thus:

- bucket equality may be required inside explicitly atomic units, such as rendezvous Push/Pop pairs and SyncChannel/ac_sync handshakes;
- bucket decomposition may be used to prove local state-transition and enablement facts;
- but CompatibleSys / $\approx_{\text{sys}}$ is defined over observable units and their mandated partial order, not over identical bucket decompositions.

3. Events, operation instances, and source order

Define dynamic operation universes:

inductive MsgKind

| Push

| Pop

| PushNB

| PopNB

def MsgKind.isBlocking : MsgKind → Bool

| MsgKind.Push => true

```

| MsgKind.Pop => true
| MsgKind.PushNB => false
| MsgKind.PopNB => false

```

```

structure DynMsgInstance where
  proc : Process
  channel : Channel
  kind : MsgKind -- Push, Pop, PushNB, PopNB
  sourceLoc : SourceLoc
  instancelId : Nat

```

```

def Blocking (q : DynMsgInstance) : Prop :=
  q.kind.isBlocking = true

```

```

def NonBlocking (q : DynMsgInstance) : Prop :=
  q.kind.isBlocking = false

```

```

structure FrontierItem where
  proc : Process
  itemKind : FrontierItemKind
  sourceLoc : SourceLoc
  instancelId : Nat

```

Core source-order objects are witness-specific. They range only over dynamic operation instances that actually occur in the compared execution / witness. Untaken branches and unexecuted loop iterations are not members of this dynamic universe.

```

Qexec : Process → Set DynMsgInstance
QOmega : Process → Set DynMsgInstance
Omega : Process → Set FrontierItem

```

Q_{exec}, P is the execution-level dynamic-instance universe. It includes executed failed non-blocking polls that may produce no Σ -event.

Ω, P is the frontier/order universe used by Front, NextFront, WFG, E3/E5, BoundaryReq attribution, and source-order reasoning. Failed NB-only executed polls are not included in Ω, P merely because they executed.

$Q_{\Omega, P}$ denotes the executed dynamic message-operation-instance slice induced by Ω, P . Since Q_{exec}, P is a set of `DynMsgInstance` and Ω, P is a set of `FrontierItem`, this is not a direct set intersection. It is defined through the message-instance projection from frontier items:

```

Q $\Omega$ ,P :=
{ q : DynMsgInstance |
  q  $\in$  Q $\text{exec}$ ,P  $\wedge$ 
   $\exists$  x : FrontierItem,
  x  $\in$   $\Omega$ ,P  $\wedge$ 
  MsgInstanceOfFrontierItem C P x = some q }

```

```

def MsgInstanceOfFrontierItem
(C : ComparisonInstance)
(P : Process)
(x : FrontierItem) : Option DynMsgInstance :=
...

```

```

structure DynamicUniverse
(C : ComparisonInstance)
(P : Process)
(W : WitnessBundle C) where
Q $\text{exec}$  : Set DynMsgInstance
 $\Omega$  : Set FrontierItem
Q_ $\Omega$  : Set DynMsgInstance

```

```

q $\Omega$ _characterization :
 $\forall$  q,
q  $\in$  Q_ $\Omega$   $\leftrightarrow$ 
q  $\in$  Q $\text{exec}$   $\wedge$ 
 $\exists$  x,
x  $\in$   $\Omega$   $\wedge$ 
MsgInstanceOfFrontierItem C P x = some q

```

```

message_frontier_items_have_exec_instance :
 $\forall$  x q,
x  $\in$   $\Omega$   $\rightarrow$ 
MsgInstanceOfFrontierItem C P x = some q  $\rightarrow$ 

```

$q \in Q_{\text{exec}} \wedge q \in Q_{\Omega}$

qomega_items_have_frontier_item :

$\forall q,$

$q \in Q_{\Omega} \rightarrow$

$\exists x,$

$x \in \Omega \wedge$

$\text{MsgInstanceOfFrontierItem } C \ P \ x = \text{some } q$

executed_only :

$\forall x,$

$x \in \Omega \rightarrow$

$\text{ExecutedInWitness } C \ W \ P \ x$

leP :

$\{x : \text{FrontierItem} \ // \ x \in \Omega\} \rightarrow$

$\{y : \text{FrontierItem} \ // \ y \in \Omega\} \rightarrow$

Prop

leP_partial_order :

PartialOrder leP

prefS :

SyncCall $\rightarrow$

Set $\{x : \text{FrontierItem} \ // \ x \in \Omega\}$

suffS :

SyncCall $\rightarrow$

Set $\{x : \text{FrontierItem} \ // \ x \in \Omega\}$

-- R2/E2 signal-anchor maps for sequential-process signal IO.

-- pred_S is defined for ordinary sequential-process signal Reads.

-- succ_S is defined for ordinary sequential-process signal Writes.

pred_S :

$\{x : \text{FrontierItem} \ // \ x \in \Omega\} \rightarrow$

Option SyncCall

succ_S :

$\{x : \text{FrontierItem} \mid x \in \Omega\} \rightarrow$
Option SyncCall

read_anchor_defined :
 $\forall x,$
SequentialSignalReadItem x.val $\rightarrow$
pred_S x $\neq$ none $\wedge$
succ_S x = none

write_anchor_defined :
 $\forall x,$
SequentialSignalWriteItem x.val $\rightarrow$
succ_S x $\neq$ none $\wedge$
pred_S x = none

pred_S_is_closest_preceding_sync :
 $\forall x s,$
SequentialSignalReadItem x.val $\rightarrow$
pred_S x = some s $\rightarrow$
ClosestPrecedingDynamicSync C P W s x.val

succ_S_is_closest_succeeding_sync :
 $\forall x s,$
SequentialSignalWriteItem x.val $\rightarrow$
succ_S x = some s $\rightarrow$
ClosestSucceedingDynamicSync C P W s x.val

signal_read_ordered_at_pred_S :
 $\forall x s,$
SequentialSignalReadItem x.val $\rightarrow$
pred_S x = some s $\rightarrow$
SignalItemClock C W x.val = SyncClock C W s

signal_write_ordered_at_succ_S :
 $\forall x s,$
SequentialSignalWriteItem x.val $\rightarrow$
succ_S x = some s $\rightarrow$
SignalItemClock C W x.val = SyncClock C W s

`MsgInstanceOfFrontierItem` is defined only for frontier items that correspond to message-operation dynamic instances. Non-message frontier items, such as pure synchronization or signal-anchor items, map to none. Signal-anchor items instead use the explicit `pred_S` / `succ_S` fields above. For sequential processes, `pred_S` places each ordinary signal Read at its closest preceding synchronization-call anchor, and `succ_S` places each ordinary signal Write at its closest succeeding synchronization-call anchor. These anchors, not raw lexical position, determine the signal item's formal position for $\leq_P$ / $<_P$, `Done_P`, `Remain_P`, `NextFront`, `E2SignalAnchorOrder`, and `CompatibleP` / `CompatibleSys` reasoning.

The field Q_Ω is therefore not a second independent universe: it is exactly the set of executed dynamic message-operation instances $q \in Q_{\text{exec}}$ for which some message frontier item $x \in \Omega$ satisfies `MsgInstanceOfFrontierItem` $C_P x = \text{some } q$. This keeps Q_{exec} , Ω , Q_Ω , and the signal-anchor universe type-distinct while making their relationships explicit for `BoundaryReq`, `Front`, `NextFront`, `WFG`, `E2`, `E3/E5` issue-order reasoning, and `R2/R2C` signal visibility.

Thus source order $\leq_P$ is not a static total order over syntactic program statements. It is a partial order over the witness-bundle-specific dynamic frontier-item universe Ω . The selector component `W.selector` determines ε -modeled source choices, while `W.nbMap` records the dynamic-instance correspondence for failed non-blocking polls that produce no Σ -event.

Non-blocking calls need explicit modeling:

- failed `PushNB` / `PopNB` polls are executed dynamic instances in Q_{exec} , but produce no Σ -event;
- successful non-blocking calls produce the corresponding committed `Push_c(v)` / `Pop_c(v)` event;
- repeated NB calls are distinct dynamic source instances even if a request signal remains physically asserted.

This matches the document's issue/commit treatment of NB calls.

4. Reactive environment and modeling contracts

4.1 Reactive environment

“Fixed stimulus” should be modeled as a causal global strategy over Σ -history, not merely as a fixed stream of per-process inputs.

```
structure ReactiveEnvironment ( $\Sigma$  : Type) where
  inputAt : SigmaHistory  $\Sigma$   $\rightarrow$  ExternalInputs
  causal :
     $\forall h_1 h_2,$ 
      SamePrefix  $h_1 h_2 \rightarrow$ 
        inputAt  $h_1$  = inputAt  $h_2$ 
```

This directly addresses the earlier issue that reactive stimulus must be interpreted as the same global Σ -causal behavior, not a function of local process history.

4.2 Modeling contracts

Direct-input contracts and array-access contracts should be parallel top-level modeling contracts, not partly folded into RRules and partly left separate.

```
structure ModelingContracts (C : ComparisonInstance) where
  directInputs : DirectInputContract C
  arrays : ArrayAccessMappingRules C
```

This keeps RRules close to the document’s actual R-rule set and treats direct-input and external-array obligations as environment/modeling contracts.

5. Direct-input contract and late sampling

The direct-input contract should contain only primitive environmental stability/update obligations. The fact that late sampling preserves the logical anchor should be a **derived theorem**, not a separate hypothesis.

The document says hls_direct_input permits late physical sampling because the environment holds the signal stable after reset, while hls_direct_input_sync permits controlled updates only at the designated synchronization handshake. The document also states that logical signal Read/Write events are timestamped at the E2 anchor, not at any later physical RTL sampling cycle; late sampling is ε -internal.

Define:

The direct-input stability rule applies only within a single reset epoch. The scheduling document explicitly permits dynamic resetting during normal HW operation, including resetting of direct-input-controlled regions.

```

def NoRelevantResetBetween
(C : ComparisonInstance)
(sig : Signal)
(t1 t2 : Cycle) : Prop :=
  ¬ ∃ r,
  t1 < r ∧
  r ≤ t2 ∧
  ReceiverResetAffectsSignal C sig r

```

```

structure DirectInputContract (C : ComparisonInstance) where
stableWithinResetEpoch :
  ∀ sig t1 t2,
  DirectInput sig →
  AfterAllReceiversOutOfReset C sig t1 →
  t1 ≤ t2 →
  NoRelevantResetBetween C sig t1 t2 →
  InputValue C.env sig t1 = InputValue C.env sig t2
syncControlledUpdateOnlyAtSync :
  ∀ sig s t,
  DirectInputSyncControlled sig s →
  ValueChangesAt C.env sig t →
  SyncHandshakeOccursAt C s t

```

Then prove:

```

theorem lateSamplingPreservesLogicalAnchor
(C : ComparisonInstance)
(hDI : DirectInputContract C)
(sig : Signal)
(anchor sampleT : Cycle)
(hWindow : DirectInputAnchorWindow C sig anchor sampleT)
(hNoReset :
  NoRelevantResetBetween C sig anchor sampleT) :
  InputValue C.env sig anchor =
  InputValue C.env sig sampleT

```

Bucket-location rule:

The logical Read(sig,val) / Write(sig,val) event belongs only to the E2 anchor bucket β_{anchor}. A physical late sample, if modeled, is an ε-step in a later bucket.

The late ε -step creates no new Σ -event. `lateSamplingPreservesLogicalAnchor` is used only to justify the value recorded in the logical anchor bucket.

structure `DirectInputBucketWF` ($C : \text{ComparisonInstance}$) where

`read_recorded_only_at_anchor :`

$\forall \text{sig val anchor sampleT } \beta,$

`DirectInput sig  $\rightarrow$`

`LogicalReadEvent sig val  $\in \text{BucketSigmaEvents } C \beta \rightarrow$`

`DirectInputAnchorWindow C sig anchor sampleT  $\rightarrow$`

`$\beta = \text{AnchorBucket } C \text{ anchor}$`

The late-sample value agreement field is reset-epoch local. It is derived from `stableWithinResetEpoch` and may be used only when no relevant dynamic reset occurs between the logical anchor cycle and the physical sample cycle.

`write_recorded_only_at_anchor :`

$\forall \text{sig val anchor } \beta,$

`DirectInput sig  $\rightarrow$`

`LogicalWriteEvent sig val  $\in \text{BucketSigmaEvents } C \beta \rightarrow$`

`$\beta = \text{AnchorBucket } C \text{ anchor}$`

`late_sample_is_epsilon_only :`

$\forall \text{sig anchor sampleT},$

`DirectInputAnchorWindow C sig anchor sampleT  $\rightarrow$`

`PhysicalLateSample C sig sampleT  $\rightarrow$`

`EpsilonStepOnly C sig sampleT  $\wedge$`

`$\neg \exists \text{val } \beta,$`

`PhysicalSampleEvent sig val sampleT  $\in \text{BucketSigmaEvents } C \beta$`

`late_sample_value_matches_anchor :`

$\forall \text{sig anchor sampleT},$

`DirectInputAnchorWindow C sig anchor sampleT  $\rightarrow$`

`NoRelevantResetBetween C sig anchor sampleT  $\rightarrow$`

`InputValue C.env sig anchor = InputValue C.env sig sampleT`

This explicitly reconciles the bucket model with direct-input late sampling:

`CompatibleP` and `CompatibleSys` compare the logical Σ -visible Read/Write events

at β_{anchor} , never the later physical sample micro-step.

6. Temporal logic layer for B2/B3/B4/B5/B6

B2 and B3 should not be opaque Prop fields. They are temporal assumptions over infinite executions.

The document defines B2 as weak fairness: if an action remains continuously enabled from some cycle onward, the scheduler must eventually select it. It applies to Push, Pop, and Sync operations. It also defines B3 as the $P_{\text{push}}(c) / P_{\text{pop}}(c)$ channel-progress obligations.

Define temporal operators:

def AlwaysFrom

(τ : InfiniteExecutionBucketed Σ)

(t : Nat)

(P : Nat $\rightarrow$ Prop) : Prop :=

$\forall n, t \leq n \rightarrow P\ n$

def EventuallyFrom

(τ : InfiniteExecutionBucketed Σ)

(t : Nat)

(P : Nat $\rightarrow$ Prop) : Prop :=

$\exists n, t \leq n \wedge P\ n$

Concrete fairness actions:

inductive FairAction

| push (q : DynMsgInstance)

| pop (q : DynMsgInstance)

| sync (s : SyncCall)

Weak fairness:

Weak fairness is a side-parametric temporal assumption. It applies to any valid infinite execution of the selected comparison side. In particular, it is not a post-only premise: the reference-side execution used by Exec_ref and the post-side execution used by RTL_B both carry the same WF/B2 obligation whenever that side is being used in a liveness/progress theorem.

def WeakFairness

```

(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma) : Prop :=
 $\forall a t,$ 
AlwaysFrom  $\tau t$  (fun n => FairActionEnabled C side  $\tau a n$ )  $\rightarrow$ 
EventuallyFrom  $\tau t$  (fun n => FairActionSelected C side  $\tau a n$ )

```

B3 channel-progress predicates should be defined in terms of persistent dynamic request instances, not per-cycle existentially chosen requests.

A blocking request is persistent from cycle t through its commit cycle inclusive when the same dynamic operation instance q keeps its BoundaryReq asserted throughout that interval. For Push operations, persistence also includes payload stability while the request is outstanding, including the commit cycle itself:

```

def PersistentRequestFrom
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(q : DynMsgInstance)
(t : Nat) : Prop :=
Blocking q  $\wedge$ 
 $\forall n,$ 
 $t \leq n \rightarrow$ 
 $\neg$  CommitsBefore C side  $\tau q n \rightarrow$ 
BoundaryReqAt C side  $\tau q n \wedge$ 
(q.kind = MsgKind.Push  $\rightarrow$ 
PayloadAt C side  $\tau q n$  = PayloadAt C side  $\tau q t$ )

```

Equivalently, one may split this into PersistentPushRequestFrom and PersistentPopRequestFrom. The single definition above is preferred here because P_push and P_pop can share the same persistence premise while Push stability is enforced only in the Push case.

```

def BufferedChannel
(C : ComparisonInstance)
(c : Channel) : Prop :=
0 < capacity C c

```

```
def RendezvousChannel
(C : ComparisonInstance)
(c : Channel) : Prop :=
capacity C c = 0
```

```
def BufferedPushStalledFullBy
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(qPush : DynMsgInstance)
(c : Channel)
(n : Nat) : Prop :=
BufferedChannel C c  $\wedge$ 
qPush.kind = MsgKind.Push  $\wedge$ 
qPush.channel = c  $\wedge$ 
Blocking qPush  $\wedge$ 
BoundaryReqAt C side  $\tau$  qPush n  $\wedge$ 
occAt c n = capacity C c
```

```
def BufferedPopStalledEmptyBy
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(qPop : DynMsgInstance)
(c : Channel)
(n : Nat) : Prop :=
BufferedChannel C c  $\wedge$ 
qPop.kind = MsgKind.Pop  $\wedge$ 
qPop.channel = c  $\wedge$ 
Blocking qPop  $\wedge$ 
BoundaryReqAt C side  $\tau$  qPop n  $\wedge$ 
occAt c n = 0
```

```
def RendezvousPairRequestingAt
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
```

```

(qPush qPop : DynMsgInstance)
(c : Channel)
(n : Nat) : Prop :=
  RendezvousChannel C c ∧
  MatchingPushRequest qPush c ∧
  MatchingPopRequest qPop c ∧
  BoundaryReqAt C side τ qPush n ∧
  BoundaryReqAt C side τ qPop n

```

```

def RendezvousProgressDischargedBy
  (C : ComparisonInstance)
  (side : Side)
  (τ : InfiniteExecutionBucketed C.Sigma)
  (qPush qPop : DynMsgInstance)
  (n : Nat) : Prop :=
  MatchingTransferCommitOn C side τ qPush qPop n

```

```

def MatchingPushRequest
  (qPush : DynMsgInstance)
  (c : Channel) : Prop :=
  qPush.kind = MsgKind.Push ∧
  qPush.channel = c ∧
  Blocking qPush

```

```

def MatchingPopRequest
  (qPop : DynMsgInstance)
  (c : Channel) : Prop :=
  qPop.kind = MsgKind.Pop ∧
  qPop.channel = c ∧
  Blocking qPop

```

```

-- Buffered full-push progress is discharged by the complementary Pop.
-- Do not count qPush itself as a direct discharge of the full condition.
-- Under E4, buffered FIFO occupancy is evaluated at the beginning of the cycle,
-- so a Push cannot commit in a cycle that begins full solely because a Pop also
-- commits in that same cycle.

```

```

def BufferedPushProgressDischargedBy
  (C : ComparisonInstance)

```

```

(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(qPush qPop : DynMsgInstance)
(n : Nat) : Prop :=
CommitAtCycle C side  $\tau$  qPop n

```

```

-- Buffered empty-pop progress is discharged by the complementary Push.
-- Do not count qPop itself as a direct discharge of the empty condition.
-- Under E4, a Pop cannot commit in a cycle that begins empty solely because a
-- Push also commits in that same cycle.

```

```

def BufferedPopProgressDischargedBy
(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(qPop qPush : DynMsgInstance)
(n : Nat) : Prop :=
CommitAtCycle C side  $\tau$  qPush n

```

```

-- P holds continuously from t until the first discharge cycle.
-- This avoids the vacuity that would result from requiring the stalled condition
-- to remain true forever after the discharge has already occurred.

```

```

def HoldsUntilDischarged
( $\tau$  : InfiniteExecutionBucketed  $\Sigma$ )
(t : Nat)
(P : Nat  $\rightarrow$  Prop)
(D : Nat  $\rightarrow$  Prop) : Prop :=
 $\forall n, t \leq n \rightarrow (\forall k, t \leq k \rightarrow k < n \rightarrow \neg D k) \rightarrow P n$ 

```

```

def EventuallyDischargedFrom
( $\tau$  : InfiniteExecutionBucketed  $\Sigma$ )
(t : Nat)
(D : Nat  $\rightarrow$  Prop) : Prop :=
 $\exists n, t \leq n \wedge D n$ 

```

Then:

```

def P_push
(C : ComparisonInstance)

```

```

(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : Channel) : Prop :=
(BufferedChannel C c  $\rightarrow$ 
 $\forall$  t qPush qPop,
MatchingPushRequest qPush c  $\rightarrow$ 
MatchingPopRequest qPop c  $\rightarrow$ 
PersistentRequestFrom C side  $\tau$  qPush t  $\rightarrow$ 
PersistentRequestFrom C side  $\tau$  qPop t  $\rightarrow$ 
BufferedPushStalledFullBy C side  $\tau$  qPush c t  $\rightarrow$ 
HoldsUntilDischarged  $\tau$  t
(fun n =>
BoundaryReqAt C side  $\tau$  qPush n  $\wedge$ 
BoundaryReqAt C side  $\tau$  qPop n  $\wedge$ 
occAt c n = capacity C c)
(fun n =>
BufferedPushProgressDischargedBy C side  $\tau$  qPush qPop n)  $\rightarrow$ 
EventuallyDischargedFrom  $\tau$  t
(fun n =>
BufferedPushProgressDischargedBy C side  $\tau$  qPush qPop n))
 $\wedge$ 
(RendezvousChannel C c  $\rightarrow$ 
 $\forall$  t qPush qPop,
MatchingPushRequest qPush c  $\rightarrow$ 
MatchingPopRequest qPop c  $\rightarrow$ 
PersistentRequestFrom C side  $\tau$  qPush t  $\rightarrow$ 
PersistentRequestFrom C side  $\tau$  qPop t  $\rightarrow$ 
RendezvousPairRequestingAt C side  $\tau$  qPush qPop c t  $\rightarrow$ 
HoldsUntilDischarged  $\tau$  t
(fun n =>
RendezvousPairRequestingAt C side  $\tau$  qPush qPop c n)
(fun n =>
RendezvousProgressDischargedBy C side  $\tau$  qPush qPop n)  $\rightarrow$ 
EventuallyDischargedFrom  $\tau$  t
(fun n =>
RendezvousProgressDischargedBy C side  $\tau$  qPush qPop n))

```

```

def P_pop

```

```

(C : ComparisonInstance)
(side : Side)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)
(c : Channel) : Prop :=
  (BufferedChannel C c  $\rightarrow$ 
     $\forall$  t qPop qPush,
      MatchingPopRequest qPop c  $\rightarrow$ 
      MatchingPushRequest qPush c  $\rightarrow$ 
      PersistentRequestFrom C side  $\tau$  qPop t  $\rightarrow$ 
      PersistentRequestFrom C side  $\tau$  qPush t  $\rightarrow$ 
      BufferedPopStalledEmptyBy C side  $\tau$  qPop c t  $\rightarrow$ 
      HoldsUntilDischarged  $\tau$  t
    (fun n =>
      BoundaryReqAt C side  $\tau$  qPop n  $\wedge$ 
      BoundaryReqAt C side  $\tau$  qPush n  $\wedge$ 
      occAt c n = 0)
    (fun n =>
      BufferedPopProgressDischargedBy C side  $\tau$  qPop qPush n)  $\rightarrow$ 
      EventuallyDischargedFrom  $\tau$  t
    (fun n =>
      BufferedPopProgressDischargedBy C side  $\tau$  qPop qPush n))
     $\wedge$ 
    (RendezvousChannel C c  $\rightarrow$ 
       $\forall$  t qPop qPush,
        MatchingPopRequest qPop c  $\rightarrow$ 
        MatchingPushRequest qPush c  $\rightarrow$ 
        PersistentRequestFrom C side  $\tau$  qPop t  $\rightarrow$ 
        PersistentRequestFrom C side  $\tau$  qPush t  $\rightarrow$ 
        RendezvousPairRequestingAt C side  $\tau$  qPush qPop c t  $\rightarrow$ 
        HoldsUntilDischarged  $\tau$  t
      (fun n =>
        RendezvousPairRequestingAt C side  $\tau$  qPush qPop c n)
      (fun n =>
        RendezvousProgressDischargedBy C side  $\tau$  qPush qPop n)  $\rightarrow$ 
        EventuallyDischargedFrom  $\tau$  t
      (fun n =>
        RendezvousProgressDischargedBy C side  $\tau$  qPush qPop n))
  )

```

This matches the document’s conditional B3 reading: B3 forces progress only when both endpoints continuously request the matching transfer; a producer stalled on a full channel is not automatically a B3 violation if the consumer is not yet requesting the matching pop.

The scheduling document assumes B1 determinism only of the post-HLS RTL_B side. The source-side models Sys / Sys_B / Sys_ref may admit multiple ε - or latency-different executions having the same Σ -projection; witness selection and snooping resolve the required source-side correspondence when needed.

For a fixed comparison instance — including the fixed initial state and fixed reactive environment / external stimulus — B1 requires uniqueness of the labeled post-side Σ -trace, not uniqueness of the full internal ε -execution. Internal ε -steps, ε -depths, and other non-observable scheduling details may differ between valid post-side executions.

```
def PostDeterministicSigmaTrace
(C : ComparisonInstance) : Prop :=
 $\forall \tau_1 \tau_2,$ 
InfinitePostExecution C  $\tau_1 \rightarrow$ 
InfinitePostExecution C  $\tau_2 \rightarrow$ 
SigmaTraceOf C .post  $\tau_1 =$ 
SigmaTraceOf C .post  $\tau_2$ 
```

Constructive ε -normalization and finite-stutter obligations for B4/B5.

B4 and B5 should not be merely opaque Prop fields when used to mechanize Lemma I.1-style arguments. They are assumptions, but they must expose enough structure to prove that internal ε -evolution cannot hide an infinite internal tail before the next Σ -action or stable wait.

This Lean strategy distinguishes two implementation-level measurements that are both silent with respect to Σ :

- intra-cycle ε -normalization: ε -microsteps inside one CycleBucket, used to reach an ε -quiescent cutpoint within a cycle; and
- cycle-advancing silent progress: one or more real clock cycles whose buckets contain no Σ -event for the process, used to model HLS-inserted FSM latency, arbitration/bookkeeping latency, failed non-blocking polls, and pipeline advances.

This distinction is only a Lean-internal decomposition. It is not a redefinition of B4. The main-document B4 premise remains a single finite ε -stutter bound for each process P: along any consecutive ε -only evolution in which P is internally advancing and is not externally stalled, P must reach either a Σ -action or a stable waiting state within a finite bound D_P .

Accordingly, ProcessEpsilonBound and IntraCycleNormalizationBound may remain separate implementation parameters, but FiniteEpsilonDepth must expose a derived unified B4 consequence. In a concrete Lean development, D_P may be chosen large enough to dominate both the cycle-advancing silent-progress bound and the intra-cycle normalization bound. The two internal bounds are useful for proof engineering, but together they must imply the single main-document B4 finite-stutter obligation.

```
def ProcessEpsilonBound
(C : ComparisonInstance)
(P : Process) : Nat :=
pipe_depth C P
```

```
def IntraCycleNormalizationBound
(C : ComparisonInstance)
(P : Process) : Nat :=
eps_norm_depth C P
```

```
-- Main-document B4 bound.
-- This is the single  $D_P$ -style finite  $\varepsilon$ -stutter bound exposed to the theorem
-- stack. The exact arithmetic expression is not important; the requirement is
-- that MainB4Bound dominate the Lean-internal cycle-progress and bucket
-- normalization bounds strongly enough to prove the unified B4 consequence.
```

```
def MainB4Bound
(C : ComparisonInstance)
(P : Process) : Nat :=
(ProcessEpsilonBound C P + 1) *
(2 * IntraCycleNormalizationBound C P + 1)
```

```
def IntraCycleEpsilonStep
(C : ComparisonInstance)
```

```

(side : Side)
( $\sigma \sigma'$  : State) : Prop :=
InternalStep C side  $\sigma \sigma'$   $\wedge$ 
SigmaEventsBetween C side  $\sigma \sigma' = \emptyset \wedge$ 
 $\sigma'.cycle = \sigma.cycle$ 

```

```

def IntraCycleEpsilonSegment
(C : ComparisonInstance)
(side : Side)
( $\sigma_0 \sigma_1$  : State)
(steps : List EpsStep) : Prop :=
EpsilonStepsFromTo C side  $\sigma_0$  steps  $\sigma_1 \wedge$ 
 $\forall \sigma \sigma'$ ,
AdjacentInEpsilonSteps C side  $\sigma_0$  steps  $\sigma_1 \sigma \sigma' \rightarrow$ 
IntraCycleEpsilonStep C side  $\sigma \sigma'$ 

```

```

def StableWaitOrSigmaEnabled
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State) : Prop :=
StableWaitCutpoint C side  $\sigma \vee$ 
 $\exists e, \text{SigmaEnabledAt } C \text{ side } e \sigma$ 

```

```

def ProcessSilentCycle
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\beta$  : CycleBucket C.Sigma) : Prop :=
WellFormedCycleBucket C side  $\beta \wedge$ 
ProcessActiveInBucket C side P  $\beta \wedge$ 
 $\beta.sigmaEvents = \emptyset \wedge$ 
InternalProgressForProcess C side P  $\beta$ 

```

```

def SilentCycleSegment
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\tau$  : InfiniteExecutionBucketed C.Sigma)

```

```

(t0 t1 : Cycle) : Prop :=
t0 ≤ t1 ∧
∀ n,
t0 ≤ n →
n < t1 →
ProcessSilentCycle C side P (τ n)

```

```

def StableWaitOrSigmaEnabledAtCycle
(C : ComparisonInstance)
(side : Side)
(P : Process)
(τ : InfiniteExecutionBucketed C.Sigma)
(n : Cycle) : Prop :=
StableWaitOrSigmaEnabled C side ((τ n).endState) ∨
∃ e,
EventOwnedByProcess C P e ∧
SigmaEnabledAt C side e ((τ n).endState)

```

```

def SigmaCommitsForProcessAt
(C : ComparisonInstance)
(side : Side)
(P : Process)
(τ : InfiniteExecutionBucketed C.Sigma)
(n : Cycle) : Prop :=
∃ e,
e ∈ (τ n).sigmaEvents ∧
EventOwnedByProcess C P e

```

```

structure FiniteEpsilonDepth (C : ComparisonInstance) where
main_B4_bound_dominates_cycle_bound :
∀ P,
ProcessEpsilonBound C P ≤ MainB4Bound C P

```

```

main_B4_bound_dominates_normalization_bound :
∀ P,
IntraCycleNormalizationBound C P ≤ MainB4Bound C P

```

-- Cycle-level consequence used by Lemma I.1-style arguments.

silent_cycle_bound :

$\forall \text{ side } \tau \in P \ t_0 \ t_1,$

InfiniteExecutionOfSide C side $\tau \rightarrow$

SilentCycleSegment C side $P \ \tau \ t_0 \ t_1 \rightarrow$

($\forall n,$

$t_0 \leq n \rightarrow$

$n < t_1 \rightarrow$

$\neg \text{StableWaitOrSigmaEnabledAtCycle C side } P \ \tau \ n) \rightarrow$

$t_1 - t_0 \leq \text{ProcessEpsilonBound C } P$

-- Main-document B4 consequence: all consecutive ε -only evolution, including

-- any intra-cycle normalization ε -steps represented inside CycleBucket, is

-- bounded by a single D_P-style bound before the process either commits a

-- Σ -action or reaches stable wait.

unified_epsilon_stutter_bound :

$\forall \text{ side } \tau \in P \ s,$

InfiniteExecutionOfSide C side $\tau \rightarrow$

ConsecutiveEpsilonOnlySegment C side $P \ \tau \ s \rightarrow$

ProcessInternallyAdvancingThroughout C side $P \ \tau \ s \rightarrow$

NoExternalStallThroughout C side $P \ \tau \ s \rightarrow$

NoSigmaCommitOrStableWaitBeforeEnd C side $P \ \tau \ s \rightarrow$

$\text{EpsilonStepCount C side } P \ \tau \ s \leq \text{MainB4Bound C } P$

normalizes_or_reaches_sigma_or_wait :

$\forall \text{ side } \tau \in P \ t_0,$

InfiniteExecutionOfSide C side $\tau \rightarrow$

ProcessActiveInBucket C side $P \ (\tau \ t_0) \rightarrow$

BeginsInternalProgressTowardNextObservable C side $P \ \tau \ t_0 \rightarrow$

$\exists n,$

$t_0 \leq n \wedge$

$n \leq t_0 + \text{MainB4Bound C } P \wedge$

($\text{SigmaCommitsForProcessAt C side } P \ \tau \ n \vee$

$\text{StableWaitOrSigmaEnabledAtCycle C side } P \ \tau \ n)$

structure QuiescentNormalization (C : ComparisonInstance) where

bucket_prefix_bounded :

$\forall \text{ side } \beta \in P,$

WellFormedCycleBucket C side $\beta \rightarrow$

ProcessOfState C side β .startState = P $\rightarrow$
 β .epsPrefix.length $\leq$ IntraCycleNormalizationBound C P

bucket_suffix_bounded :
 $\forall$ side β P,
WellFormedCycleBucket C side $\beta \rightarrow$
ProcessOfState C side β .endState = P $\rightarrow$
 β .epsSuffix.length $\leq$ IntraCycleNormalizationBound C P

end_quiescent_sound :
 $\forall$ side β ,
WellFormedCycleBucket C side $\beta \rightarrow$
 β .endQuiescent $\rightarrow$
QuiescentAt C side β .endState

system_quiescence_bound :
Nat

system_quiescence_bound_is_max :
 $\forall$ P,
ProcessEpsilonBound C P $\leq$ system_quiescence_bound

no_epsilon_steps_in_quiescent_bucket :
 $\forall$ side β ,
WellFormedCycleBucket C side $\beta \rightarrow$
QuiescentAt C side β .startState $\rightarrow$
 β .epsPrefix = [] $\wedge$ β .epsSuffix = [] $\wedge$ β .endState = β .startState

bounded_system_quiescence :
 $\forall$ side τ t,
InfiniteExecutionOfSide C side $\tau \rightarrow$
NoObservableActionEnabledFrom C side τ t $\rightarrow$
 $\exists$ n,
t $\leq$ n $\wedge$
n $\leq$ t + system_quiescence_bound $\wedge$
QuiescentAt C side ((τ n).endState) $\wedge$
 $\forall$ m,
n $\leq$ m $\rightarrow$

QuiescentAt C side $((\tau \ m).startState) \wedge$
 QuiescentAt C side $((\tau \ m).endState) \wedge$
 $(\tau \ m).epsPrefix = [] \wedge$
 $(\tau \ m).epsSuffix = []$

The bounded_system_quiescence field is the Lean-level form of B5: once no observable action is enabled anywhere, the system must reach the ε -fixed point within the explicit bound $\max_P \ D_P$, represented here by system_quiescence_bound, and no later bucket may contain further ε -microsteps unless an external change invalidates NoObservableActionEnabledFrom.

B4_finiteEpsilonDepth exposes the main-document B4 consequence through MainB4Bound: each process has a single finite D_P -style ε -stutter bound before a Σ -action or stable wait. ProcessEpsilonBound and IntraCycleNormalizationBound remain useful Lean-internal components, but they are not separate replacements for B4. The formalization must prove that the internal decomposition implies the single unified B4 finite-stutter obligation. QuiescentNormalization may still use IntraCycleNormalizationBound for bucket well-formedness and ε -quiescent cutpoint construction.

Bundle B-rules:
 structure BRules (C : ComparisonInstance) where
 B1_postDeterminism :
 PostDeterministicSigmaTrace C

-- B2 is not post-only. It applies to any valid infinite execution of the
 -- selected comparison side when that side participates in a progress/liveness
 -- theorem.

B2_weakFairness :
 $\forall$ side τ ,
 InfiniteExecutionOfSide C side $\tau \rightarrow$
 WeakFairness C side τ

-- B3 is likewise side-parametric. The P_push/P_pop obligations are required
 -- for the reference-side Sys_ref execution as well as the post-side RTL_B
 -- execution when the theorem uses those executions for liveness/deadlock
 -- reasoning.

B3_channelProgress :

$\forall \text{ side } \tau c,$
 $\text{InfiniteExecutionOfSide } C \text{ side } \tau \rightarrow$
 $P_{\text{push}} C \text{ side } \tau c \wedge P_{\text{pop}} C \text{ side } \tau c$

$B4_{\text{finiteEpsilonDepth}} :$
 $\text{FiniteEpsilonDepth } C$
 $B5_{\text{quiescentNormalization}} :$
 $\text{QuiescentNormalization } C$
 $B6_{\text{idleStutterTimeDivergence}} :$
 $\text{IdleStutterTimeDivergence } C$

$B2/B3$ remain temporal assumptions usable by K proofs. $B4/B5$ also remain assumptions, but they are now constructive assumptions: they provide the finite ε -normalization facts needed to bound epsPrefix / epsSuffix lengths in CycleBucket and to prove bounded ε -chain lemmas such as Lemma I.1.

7. Channel semantics and E4

Define channel state over the selected Sys_ref boundary.

The primitive abstract occupancy is cycle-boundary, non-fall-through occupancy. It is indexed by the cycle, not by the full intra-cycle state:

$\text{capacity} : C.\text{Sys_ref}.\text{Channel} \rightarrow \text{Nat}$
 $\text{occAt} : C.\text{Sys_ref}.\text{Channel} \rightarrow \text{Cycle} \rightarrow \text{Nat}$

For convenience, one may define a state-indexed abbreviation only by projecting the state to its cycle:

$\text{occAtState} : C.\text{Sys_ref}.\text{Channel} \rightarrow \text{State} \rightarrow \text{Nat}$
 $\text{occAtState } c \sigma := \text{occAt } c \sigma.\text{cycle}$

This abbreviation is not a second occupancy notion. In particular, occupancy is invariant under ε -steps and under same-cycle Σ -events:

$\text{occ_invariant_within_cycle} :$
 $\forall c \sigma_1 \sigma_2,$
 $\sigma_1.\text{cycle} = \sigma_2.\text{cycle} \rightarrow$

$\text{occAtState } c \ \sigma_1 = \text{occAtState } c \ \sigma_2$

-- Domain for the one-cycle Appendix C request-at-issue obligation.
-- This domain includes every executed source-level message-operation instance,
-- including failed PushNB / PopNB polls that produce no committed Σ -event.

def BoundaryReqIssueDomainSide

(C : ComparisonInstance)

(side : Side)

(q : DynMsgInstance)

(σ : State) : Prop :=

$\exists P,$

$q \in Q\text{ExecSide } C \text{ side } P \wedge$

$\text{MessageOperationInstance } q \wedge$

$\text{OwnsMessageEndpoint } P \ q \wedge$

$\text{IssuedAtState } C \text{ side } q \ \sigma$

-- Domain for persistent operation-attributive request/channel reasoning.
-- This is the domain used by ChannelEnabled, ChannelDisabled, Front, NextFront,
-- automatic flush, and WFG construction. Failed NB-only polls are not included
-- merely because they executed.

def BoundaryReqPersistentDomainSide

(C : ComparisonInstance)

(side : Side)

(q : DynMsgInstance)

(σ : State) : Prop :=

$\exists P,$

$q \in Q\Omega\text{Side } C \text{ side } P \wedge$

$\text{MessageOperationInstance } q \wedge$

$\text{OwnsMessageEndpoint } P \ q$

-- General BoundaryReq domain.
-- A request may be meaningful either because q is issuing in the current cycle
-- or because q is a persistent message-operation frontier instance.

def BoundaryReqDomainSide

(C : ComparisonInstance)

(side : Side)

(q : DynMsgInstance)

(σ : State) : Prop :=

BoundaryReqIssueDomainSide C side q $\sigma \vee$
BoundaryReqPersistentDomainSide C side q σ

Executed failed non-blocking polls may belong to Q_{exec}, P for replay and witness purposes while remaining absent from Q_{Ω}, P and Ω_P . Such polls do acquire the one-cycle BoundaryReq-at-issue obligation when they issue, via BoundaryReqIssueDomainSide. They do not acquire persistent BoundaryReqPersistentDomainSide, ChannelEnabled, ChannelDisabled, Front, NextFront, automatic-flush, or WFG obligations merely because they executed.

BoundaryReqSide :
ComparisonInstance $\rightarrow$ Side $\rightarrow$ DynMsgInstance $\rightarrow$ State $\rightarrow$ Prop

ChannelEnabledSide :
ComparisonInstance $\rightarrow$ Side $\rightarrow$ DynMsgInstance $\rightarrow$ State $\rightarrow$ Prop

```
def ChannelDisabledSide
(C : ComparisonInstance)
(side : Side)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
BoundaryReqPersistentDomainSide C side q  $\sigma \wedge$ 
BoundaryReqSide C side q  $\sigma \wedge$ 
 $\neg$  ChannelEnabledSide C side q  $\sigma$ 
```

For post-side automatic-flush and issue/commit formulas, use the following abbreviations:

```
def BoundaryReqIssueDomain
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
BoundaryReqIssueDomainSide C .post q  $\sigma$ 
```

```
def BoundaryReqPersistentDomain
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
```

BoundaryReqPersistentDomainSide C .post q σ

```
def BoundaryReqDomain
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
BoundaryReqDomainSide C .post q  $\sigma$ 
```

```
def BoundaryReq
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
BoundaryReqSide C .post q  $\sigma$ 
```

```
def ChannelEnabled
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
ChannelEnabledSide C .post q  $\sigma$ 
```

```
def ChannelDisabled
(C : ComparisonInstance)
(q : DynMsgInstance)
( $\sigma$  : State) : Prop :=
ChannelDisabledSide C .post q  $\sigma$ 
```

BoundaryReqSide, ChannelEnabledSide, and ChannelDisabledSide are the canonical side-indexed predicates. The shorter BoundaryReq / ChannelEnabled / ChannelDisabled forms are post-side abbreviations used only where the section is already explicitly about RTL_B / post-side behavior.

BoundaryReqSide is used under BoundaryReqDomainSide. This includes both one-cycle issue requests, via BoundaryReqIssueDomainSide, and persistent frontier/channel requests, via BoundaryReqPersistentDomainSide.

ChannelEnabledSide and ChannelDisabledSide are used only under BoundaryReqPersistentDomainSide or its post-side abbreviation BoundaryReqPersistentDomain. Thus failed non-blocking polls in

$Q_exec, P \setminus \Omega_P$ may satisfy the Appendix C one-cycle request-at-issue rule, but they still do not become message-operation frontier items and do not acquire ChannelEnabled, ChannelDisabled, Front, NextFront, automatic-flush, or WFG obligations merely because they executed.

ChannelEnabled is allowed to inspect the settled boundary-request state σ for persistent message-operation frontier instances, but its capacity/occupancy component must use $occAt\ q.channel\ \sigma.cycle$. Thus E4 enablement is evaluated against the beginning-of-cycle abstract occupancy, with no same-cycle fall-through through ε -settling or same-cycle Σ -events.

Capacity cases:

inductive CapacityKind
| rendezvous -- $B(c) = 0$
| buffered -- $B(c) > 0$

Expected lemmas:

BufferedPushEnabled
BufferedPopEnabled
RendezvousEnabled
FIFOOrderPreservation
NoDropNoDuplicate
OccupancyUpdatePush
OccupancyUpdatePop

E4 should be parameterized by Sys_ref, so in elaborated cases it ranges over explicit elaborated channels, not merely outer channels.

8. Appendix Q elaboration interface

Appendix Q should be introduced early. In the updated document, elaboration is not a late proof afterthought: in elaborated cases, $Sys_ref = Sys_B^{elab}(E)$, and all references to B, occ, BoundaryReq, ChannelEnabled/Disabled, Front_P, and WFG are interpreted over the explicit channels of the elaborated model.

As in the ordinary E4 semantics, elaborated-channel occupancy is cycle-boundary, non-fall-through occupancy. Thus $occ_E(d,t)$ and $A_E(c,t)$ are indexed by cycle t . State-indexed forms such as $occ_E(d,\sigma)$ or $A_E(c,\sigma)$, when used informally at ε -quiescent cutpoints, are

only abbreviations for $\text{occ_E}(d, \sigma.\text{cycle})$ and $\text{A_E}(c, \sigma.\text{cycle})$.

Chan_outer should be a parameter fixed by the chosen outer Sys_B , not a field chosen internally by the elaboration package.

The elaboration projection is intentionally multi-valued. A proof-internal elaborated channel may contribute to no outer channel, one outer channel, or several outer channels. Separately, a proof-internal token/event may account for multiple outer Σ_{DUT} -visible channel effects in bundled or joined cases.

```
structure ElaborationPackage
```

```
(Sys_B : System)
```

```
(Chan_outer : Type)
```

```
( $\Sigma_{\text{DUT}}$  : Type) where
```

```
Sys_elab : System
```

```
Chan_elab : Type
```

```
finite_chan_elab : Fintype Chan_elab
```

```
ElabToken : Type
```

```
B_E : Chan_elab  $\rightarrow$  Nat
```

```
occ_E : Chan_elab  $\rightarrow$  Cycle  $\rightarrow$  Nat
```

```
-- Channel-level accounting support.
```

```
--  $\Pi_{\text{elab}} d$  is the set of outer channels whose aggregate accounting may depend
```

```
-- on elaborated channel  $d$ . The empty set represents a proof-internal channel
```

```
-- that is hidden from the outer DUT/testbench boundary.
```

```
Pi_elab : Chan_elab  $\rightarrow$  Set Chan_outer
```

```
-- Token/event-level accounting support.
```

```
--  $\Pi_{\text{token\_elab}} \text{tok}$  is the set of outer channels accounted for by one
```

```
-- proof-internal token/event. This may contain multiple outer channels for
```

```
-- bundled or joined work items.
```

```
Pi_token_elab : ElabToken  $\rightarrow$  Set Chan_outer
```

```
tokenOfSigma : Sys_elab.Sigma  $\rightarrow$  Option ElabToken
```

```
piSigma : Trace Sys_elab.Sigma  $\rightarrow$  Trace  $\Sigma_{\text{DUT}}$ 
```

```
A_E : Chan_outer  $\rightarrow$  Cycle  $\rightarrow$  Nat
```

```
outer_trace_preservation : Prop
```

```
occupancy_preservation : Prop
```

`internal_refinement : Prop`
`no_hidden_cross_channel_disablement : Prop`
`wfg_visibility : Prop`
`combinational_wellformedness : Prop`

For each outer channel c , the channel-level fiber of the elaboration is now understood as

$$\Pi_{\text{elab}}^{-1}(c) = \{ d : \text{Chan_elab} \mid c \in \text{Pi_elab } d \}.$$

This replaces the single-valued Option-based fiber. The token-level projection Pi_token_elab is the more precise object used when one proof-internal staged/bundle/join token accounts for several outer Σ_{DUT} -visible channel effects. The trace projection piSigma and the occupancy/accounting map A_E must be consistent with both the channel-level support map Pi_elab and the token-level projection Pi_token_elab .

For convenience, the elaboration package may also define state-indexed abbreviations by projecting the state to its cycle:

`occ_E_atState : Chan_elab → State → Nat`
`occ_E_atState d σ := occ_E d σ.cycle`

`A_E_atState : Chan_outer → State → Nat`
`A_E_atState c σ := A_E c σ.cycle`

These abbreviations are not separate occupancy or accounting notions. They are invariant under ε -steps and same-cycle Σ -events:

`occ_E_invariant_within_cycle :`
 $\forall d \sigma_1 \sigma_2,$
 $\sigma_1.\text{cycle} = \sigma_2.\text{cycle} \rightarrow$
`occ_E_atState d σ1 = occ_E_atState d σ2`

`A_E_invariant_within_cycle :`
 $\forall c \sigma_1 \sigma_2,$
 $\sigma_1.\text{cycle} = \sigma_2.\text{cycle} \rightarrow$
`A_E_atState c σ1 = A_E_atState c σ2`

Reference choice:

inductive ReferenceKind

| ordinarySysB

| elaborated

structure SysRefChoice where

outerSysB : System

Chan_outer : Type

kind : ReferenceKind

elab? : Option (ElaborationPackage outerSysB Chan_outer Σ _DUT)

Sys_ref : System

sys_ref_matches_kind : Prop

For ordinarySysB, Sys_ref = outerSysB.

For elaborated, Sys_ref = E.Sys_elab.

This also handles sync-boundary epoch gates: if proof-relevant withholding exists, it must be represented by explicit gate channels in $\text{Sys_B}^{\text{elab}}(E)$, not by hidden gating on an otherwise E4-enabled outer channel. The earlier review correctly flagged this as a point that must be explicit in the Lean strategy.

Locality note. Although ElaborationPackage is presented as a single structure parameterizing the theorem instance, its components (Sys_elab, Chan_elab, Π _elab, Π _token,elab, π _ Σ ,E, A_E) are constructed locally per process and per channel boundary. The channel-level projection Π _elab is multi-valued, and the token-level projection Π _token,elab is allowed to map one proof-internal staged/bundle/join token to multiple outer Σ _DUT-visible channels. Each per-process elaboration is determined by that process's local channel structure (number of inputs, sync-boundary discipline, pipelining configuration) and is independent of the elaboration chosen for any other process. The system-level ElaborationPackage is the disjoint composition of these per-process pieces. Treating E as a theorem-instance parameter is the right Lean encoding of "the verifier has chosen per-process elaborations and the proof takes them as given"; it is not a claim that the verifier must reason globally to construct E.

9. R-rules, E-rules, and modeling contracts

The R-rule bundle should include only the document's actual R-rules plus an optional Lean-side array-rule pointer if desired. It should not include a fictional

R5_syncBoundaryNoCrossing; sync-boundary crossing is E5. The document explicitly labels “Messages cannot cross their immediate synchronization boundary” as E5.

However, R5 itself is a real R-rule: R5 (Channel capacity semantics; Sys vs. Sys_B). In the Lean strategy, R5 should record the chosen source/reference-model capacity convention: raw Sys uses rendezvous capacity, while Sys_B / Sys_ref uses the selected capacity B(c), with B(c)=0 interpreted as rendezvous and B(c)>0 interpreted as cycle-boundary, non-fall-through bounded FIFO. The concrete enablement and occupancy-update obligations remain the E4 channel-capacity semantics; R5 records which source/reference capacity semantics are being used by the selected comparison instance.

```
def R2CCombinationalFixedPointRule
(C : ComparisonInstance) : Prop :=
SequentialOnly C ∨
∃ hR2C : R2C_WF C,
R2CEventsObservedOnlyAtFixedPoint C hR2C
```

```
def R5ChannelCapacitySemantics
(C : ComparisonInstance) : Prop :=
SysRefUsesSelectedCapacitySemantics C ∧
∀ c,
(capacity C c = 0 → RendezvousChannel C c) ∧
(0 < capacity C c → BufferedChannel C c)
```

```
structure RRules (C : ComparisonInstance) where
R1_syncOrder : Prop
R2_signalAnchorSequential : Prop
R2C_combinationalFixedPoint :
R2CCombinationalFixedPointRule C
R3_syncSemantics : Prop
R4_messageIssueOrder : Prop
R5_channelCapacitySemantics :
R5ChannelCapacitySemantics C
```

R4_messageIssueOrder is deliberately named only as an issue-order rule. It records the source-induced no-reverse discipline for message-operation issue.

It does not impose a separate commit-order rule; commit timing is constrained by E4 channel availability, FIFO/rendezvous semantics, backpressure, and the chosen Sys_ref / RTL_B comparison instance.

E-rules:

structure ERules (C : ComparisonInstance) where

E1_sync_order_preservation : Prop

E2_signal_anchor_preservation : Prop

E3_message_order_preservation : Prop

E4_channel_capacity_semantics : Prop

E5_sync_boundary_preservation : Prop

Array and direct-input conditions live in:

structure ModelingContracts (C : ComparisonInstance) where

directInputs : DirectInputContract C

arrays : ArrayAccessMappingRules C

This avoids treating environment/design-side contracts inconsistently.

10. Array access mapping

External arrays are not merely internal Store. The document states that external arrays are mapped onto IO operations by an array access mapping layer, and that array accesses before a synchronization call must commit no later than the sync commit, while accesses after the sync must commit strictly after it. It also permits transformations such as caching, merging, splitting, and reordering when allowed by the mapping layer.

Define:

inductive MemoryEvent

| read (array : ArrayId) (addr : Addr) (value : Value)

| write (array : ArrayId) (addr : Addr) (value : Value)

structure ArrayAccessMapping where

sourceArrayAccess : SourceArrayAccess → List CoreIOEvent

commitTime : SourceArrayAccess → Cycle

fixedLatency : SourceArrayAccess → Nat

conflictFreeReorderAllowed :

$\text{SourceArrayAccess} \rightarrow \text{SourceArrayAccess} \rightarrow \text{Prop}$

transformedEquivalent :

$\text{SourceArrayAccess} \rightarrow \text{List SourceArrayAccess} \rightarrow \text{Prop}$

Rules:

structure ArrayAccessMappingRules (C : ComparisonInstance) where

beforeSyncCommitsNoLater :

$\forall a\ s,$

$a \in \text{prefS } s \rightarrow$

$\text{ArrayCommitTime } a \leq \text{SyncCommitTime } s$

afterSyncCommitsAfter :

$\forall a\ s,$

$a \in \text{suffS } s \rightarrow$

$\text{SyncCommitTime } s < \text{ArrayCommitTime } a$

issueConsistentWithFixedLatency :

$\forall a,$

$\text{ArrayCommitTime } a =$

$\text{ArrayIssueTime } a + \text{ArrayFixedLatency } a$

transformedOpsRespectMapping :

$\forall a\ \text{transformed},$

$\text{TransformedFrom } a\ \text{transformed} \rightarrow$

$\text{MappingEquivalent } a\ \text{transformed}$

conflictFreeReorderingOnly :

$\forall a_1\ a_2,$

$\text{Reordered } a_1\ a_2 \rightarrow$

$\text{conflictFreeReorderAllowed } a_1\ a_2$

For early milestones, it is acceptable to assume:

AllExternalArrayAccessesAlreadyMappedToCoreIO C

but the full proof strategy should include array mapping.

11. R2C combinational fixed point

ComparisonInstance now has:

combNetwork? : Option CombNetwork

R2C should use that field when combinational processes are in scope. The document says combinational Read/Write events occur in the observable bucket corresponding to the cycle's ε -quiescent combinational fixed point, and that combinational signal IO is well formed only when the relevant network is deterministic and reaches a unique ε -fixed point.

Define:

structure CombNetwork where
combStep : Store $\rightarrow$ Store $\rightarrow$ Prop
deterministic : Prop

The scheduling document's R2C rule requires deterministic ε -settling to a unique observable fixed point. It does not globally require structural combinational acyclicity for every admissible combinational network. Pure combinational oscillation, ambiguous multi-driver resolution, or non-convergent ε -behavior are excluded instead through the unique-fixed-point obligation below.

Structural acyclicity may still be imposed as a separate well-formedness requirement for specific elaboration/coupler constructions in Appendix Q, but it is not a universal requirement of CombNetwork itself.

def CombQuiescent (N : CombNetwork) (σ : State) : Prop :=
 $\neg \exists \sigma', N.\text{combStep } \sigma.\text{store } \sigma'.\text{store}$

def CombFixedPointAtCycle
 (N : CombNetwork)
 (t : Cycle)
 (μ : Store) : Prop :=
 ReachesByCombSteps N t μ $\wedge$
 IsFixedPoint N μ

Well-formedness:

structure R2C_WF (C : ComparisonInstance) where
network_present :
C.combNetwork?.isSome
uniqueFixedPoint :

```

 $\forall t \ N,$ 
 $C.\text{combNetwork?} = \text{some } N \rightarrow$ 
 $\exists! \mu, \text{CombFixedPointAtCycle } N \ t \ \mu$ 
 $\text{sigmaReadsAtFixedPoint} :$ 
 $\forall e \ t \ \mu \ N,$ 
 $C.\text{combNetwork?} = \text{some } N \rightarrow$ 
 $\text{CombReadWriteEvent } e \rightarrow$ 
 $e \in \text{BucketSigmaEvents } C \ t \rightarrow$ 
 $\text{CombFixedPointAtCycle } N \ t \ \mu \rightarrow$ 
 $\text{EventReadsWritesStore } e \ \mu$ 

def R2CEventsObservedOnlyAtFixedPoint
(C : ComparisonInstance)
(hR2C : R2C_WF C) : Prop :=
 $\forall e \ t,$ 
 $\text{CombReadWriteEvent } e \rightarrow$ 
 $e \in \text{BucketSigmaEvents } C \ t \rightarrow$ 
 $\exists N \ \mu,$ 
 $C.\text{combNetwork?} = \text{some } N \wedge$ 
 $\text{CombFixedPointAtCycle } N \ t \ \mu \wedge$ 
 $\text{EventReadsWritesStore } e \ \mu$ 

```

If combinational processes are out of scope for an early milestone, use:

SequentialOnly C

and postpone R2C_WF.

12. Issue/Commit well-formedness

Issue and Commit should be relational plus existence/uniqueness, not necessarily computable total functions.

IssueAt : Trace C.Sys_ref $\rightarrow$ DynMsgInstance $\rightarrow$ Cycle $\rightarrow$ Prop

CommitAt : Trace C.Sys_ref $\rightarrow$ DynMsgInstance $\rightarrow$ Cycle $\rightarrow$ Payload $\rightarrow$ Prop

Well-formedness:

```

def BoundaryRequestAtIssue
(C : ComparisonInstance) : Prop :=
 $\forall \tau \ q \ t \ \text{Issue } \sigma,$ 

```

IssuedInTrace $C \tau q \rightarrow$
 IssueAt $\tau q \text{ tIssue} \rightarrow$
 StateAtCycle $\tau \text{ tIssue} \sigma \rightarrow$
 BoundaryReqIssueDomain $C q \sigma \wedge$
 BoundaryReq $C q \sigma \wedge$
 $((q.\text{kind} = \text{MsgKind.Push} \vee q.\text{kind} = \text{MsgKind.PushNB}) \rightarrow$
 PayloadAtState $C q \sigma = \text{PayloadAtIssue } C \tau q \text{ tIssue})$

def BlockingBoundaryRequestPersistence
 ($C : \text{ComparisonInstance}$) : Prop :=
 $\forall \tau q \text{ tIssue } t \sigma,$
 IssuedInTrace $C \tau q \rightarrow$
 IssueAt $\tau q \text{ tIssue} \rightarrow$
 StateAtCycle $\tau t \sigma \rightarrow$
 $\text{tIssue} \leq t \rightarrow$
 $\neg \text{CommitsBefore } C \tau q t \rightarrow$
 Blocking $q \rightarrow$
 BoundaryReqPersistentDomain $C q \sigma \wedge$
 BoundaryReq $C q \sigma \wedge$
 $(q.\text{kind} = \text{MsgKind.Push} \rightarrow$
 PayloadAtState $C q \sigma = \text{PayloadAtIssue } C \tau q \text{ tIssue})$

def BoundaryRequestSoundness
 ($C : \text{ComparisonInstance}$) : Prop :=
 BoundaryRequestAtIssue $C \wedge$
 BlockingBoundaryRequestPersistence C

def StableAttribution
 ($C : \text{ComparisonInstance}$) : Prop :=
 $\forall \tau q_0 q_1 t \sigma t \sigma_{\text{next}},$
 SameEndpointDirection $q_0 q_1 \rightarrow$
 $q_0 \neq q_1 \rightarrow$
 Blocking $q_0 \rightarrow$
 StateAtCycle $\tau t \sigma \rightarrow$
 StateAtCycle $\tau (t + 1) \sigma_{\text{next}} \rightarrow$
 BoundaryReqDomain $C q_0 \sigma \rightarrow$
 BoundaryReq $C q_0 \sigma \rightarrow$
 BoundaryReqDomain $C q_1 \sigma_{\text{next}} \rightarrow$

BoundaryReq C q_1 $\sigma_{next} \rightarrow$
 $\neg \text{EndpointCommitAt } C \tau q_0.\text{endpoint } t \rightarrow$
 $\neg \text{EndpointCommitAt } C \tau q_1.\text{endpoint } t \rightarrow$
 False

BoundaryRequestSoundness has two parts. BoundaryRequestAtIssue applies to every issued source-level message-operation instance q , including PushNB and PopNB instances that fail and therefore produce no committed Σ -event. Such a non-blocking poll must still assert the appropriate endpoint request in its issue cycle. BlockingBoundaryRequestPersistence is the additional non-withdrawal rule for blocking Push/Pop instances: after issue and before commit, the same dynamic operation instance keeps its request asserted, and a Push keeps its payload stable.

StableAttribution applies when the earlier outstanding request q_0 is a blocking Push/Pop. In that case, attribution cannot change to any distinct same-endpoint operation before a commit on that endpoint direction, whether the later candidate q_1 is blocking or non-blocking. For non-blocking PushNB / PopNB calls, request assertion at the issue cycle does not by itself imply persistence beyond that issue cycle, and continuous assertion of the endpoint request signal does not by itself collapse later executed non-blocking call instances into the same dynamic instance q . Repeated executed NB polls remain distinct source-level dynamic instances unless the witness explicitly identifies the same still-outstanding request instance.

-- Dynamic source interval between synchronization anchors.
 -- This is witness-specific: only executed dynamic instances in the selected
 -- witness universe are classified. Untaken branches and unexecuted loop
 -- iterations have no SyncIntervalId.

structure SyncInterval where
 proc : Process
 pred? : Option SyncCall
 succ? : Option SyncCall
 intervalId : Nat

def SyncIntervalId

```

(C : ComparisonInstance)
(W : WitnessBundle C)
(q : DynMsgInstance) : Option SyncInterval :=
DynamicSourceIntervalOf C W q

```

```

structure SyncIntervalWF
(C : ComparisonInstance)
(W : WitnessBundle C) where
interval_defined_for_frontier_messages :
 $\forall P q,$ 
 $q \in (C.witness.dynamicUniverse P W).Q\_ \Omega \rightarrow$ 
 $\exists I, SyncIntervalId C W q = some I$ 

```

```

same_interval_iff_same_anchors :
 $\forall q_0 q_1 I_0 I_1,$ 
SyncIntervalId C W  $q_0 = some I_0 \rightarrow$ 
SyncIntervalId C W  $q_1 = some I_1 \rightarrow$ 
 $I_0 = I_1 \leftrightarrow$ 
 $I_0.proc = I_1.proc \wedge$ 
 $I_0.pred? = I_1.pred? \wedge$ 
 $I_0.succ? = I_1.succ? \wedge$ 
 $I_0.intervalId = I_1.intervalId$ 

```

```

pref_suff_consistent :
 $\forall s q,$ 
 $q \in prefS s \vee q \in suffS s \rightarrow$ 
 $\exists I, SyncIntervalId C W q = some I$ 

```

```

def SameSyncInterval
(C : ComparisonInstance)
(W : WitnessBundle C)
( $q_0 q_1 : DynMsgInstance$ ) : Prop :=
 $\exists I,$ 
SyncIntervalId C W  $q_0 = some I \wedge$ 
SyncIntervalId C W  $q_1 = some I$ 

```

```

def BackToBackIssueWF

```

```

(C : ComparisonInstance)
(W : WitnessBundle C) : Prop :=
  ∀ τ q0 q1 t,
  SameEndpointDirection q0 q1 →
  SameSyncInterval C W q0 q1 →
  CommitAtCycle C τ q0 t →
  NextSourceMessageOpOnEndpoint C q0 q1 →
  EndpointRequestContinuouslyAsserted C τ q0.endpoint t (t + 1) →
  IssueAt τ q1 (t + 1)

```

The SameSyncInterval premise prevents the back-to-back issuance rule from carrying a continuously asserted endpoint request across an explicit wait(), SyncChannel, ac_sync, or other synchronization boundary. Cross-boundary issue timing is governed instead by SyncBoundaryIssueDiscipline / E5.

```

def SyncBoundaryIssueDiscipline
(C : ComparisonInstance) : Prop :=
  ∀ τ q0 q1 s tSync t0 t1,
  q0 ∈ prefS s →
  q1 ∈ suffS s →
  SyncCommitAt C τ s tSync →
  IssueAt τ q0 t0 →
  IssueAt τ q1 t1 →
  t0 ≤ tSync ∧ tSync < t1

```

```

def NBDynamicInstanceDiscipline
(C : ComparisonInstance) : Prop :=
  ∀ P q0 q1,
  IsNB q0 →
  IsNB q1 →
  ExecutedInProcess C P q0 →
  ExecutedInProcess C P q1 →
  SameSourceCallSiteButDifferentExecution q0 q1 →
  q0 ≠ q1

```

```

structure IssueCommitWF
(C : ComparisonInstance)

```

$(W : \text{WitnessBundle } C)$ where
 $\text{issue_exists} :$
 $\forall \tau q,$
 $\text{IssuedInTrace } C \tau q \rightarrow$
 $\exists t, \text{IssueAt } \tau q t$
 $\text{issue_unique} :$
 $\forall \tau q t_1 t_2,$
 $\text{IssueAt } \tau q t_1 \rightarrow$
 $\text{IssueAt } \tau q t_2 \rightarrow$
 $t_1 = t_2$
 $\text{commit_exists_unique} :$
 $\forall \tau q,$
 $\text{CommitsInTrace } C \tau q \rightarrow$
 $\exists! tp : \text{Cycle} \times \text{Payload},$
 $\text{CommitAt } \tau q tp.1 tp.2$
 $\text{boundary_request_soundness} :$
 $\text{BoundaryRequestSoundness } C$
 $\text{stable_attribution} :$
 $\text{StableAttribution } C$
 $\text{back_to_back_issue} :$
 $\text{BackToBackIssueWF } C W$
 $\text{sync_boundary_discipline} :$
 $\text{SyncBoundaryIssueDiscipline } C$
 $\text{nb_dynamic_instance_discipline} :$
 $\text{NBDynamicInstanceDiscipline } C$

Derived timestamp:
 noncomputable def IssueTime
 $(hWF : \text{IssueCommitWF } C W)$
 $(h : \text{IssuedInTrace } C \tau q) : \text{Cycle} :=$
 $\text{Classical.choose } (hWF.\text{issue_exists } \tau q h)$

This preserves Appendix C's use of $\text{Issue}(q)$ as an auxiliary timestamp while avoiding an unnecessary computability requirement.

13. Automatic flush and sync-boundary policy

Define:

Pending $P \ \sigma$

Front $P \ \sigma$

NextFront $P \ \sigma$

BlockingMsgNextFront $P \ \sigma$

Done $P \ \sigma$

Remain $P \ \sigma$

Automatic flush must be stated over the same dynamic universe and source-order objects used by Appendix I/J/K. In particular, it must not be an opaque Prop bundle.

def FrontierBlocked

(C : ComparisonInstance)

(P : Process)

(σ : State) : Prop :=

Front C P $\sigma \neq \emptyset \wedge$

$\forall x,$

$x \in \text{Front C P } \sigma \rightarrow$

BlockingMessageFrontierItem x $\wedge$

BoundaryReq C x $\sigma \wedge$

ChannelDisabled C x σ

def LaterThanBlockedFrontier

(C : ComparisonInstance)

(P : Process)

(σ : State)

(y : FrontierItem) : Prop :=

$\exists x,$

$x \in \text{Front C P } \sigma \wedge$

SourceOrderLt C P x y

def FrontierRemainsBlockedOver

(C : ComparisonInstance)

(P : Process)

(τ : Trace C.RTL_B)

$(t_0 \ t : \text{Cycle}) : \text{Prop} :=$
 $\forall n \ \sigma n,$
 $t_0 \leq n \rightarrow$
 $n \leq t \rightarrow$
 $\text{StateAtCycle } \tau \ n \ \sigma n \rightarrow$
 $\text{FrontierBlocked } C \ P \ \sigma n$

$\text{def NoNewLaterWorkWhileBlocked}$
 $(C : \text{ComparisonInstance})$
 $(P : \text{Process})$
 $(\tau : \text{Trace } C.\text{RTL_B}) : \text{Prop} :=$
 $\forall t_0 \ t \ \sigma_0 \ y,$
 $\text{StateAtCycle } \tau \ t_0 \ \sigma_0 \rightarrow$
 $\text{FrontierBlocked } C \ P \ \sigma_0 \rightarrow$
 $\text{FrontierRemainsBlockedOver } C \ P \ \tau \ t_0 \ t \rightarrow$
 $\text{LaterThanBlockedFrontier } C \ P \ \sigma_0 \ y \rightarrow$
 $\neg \text{IssueAt } \tau \ y \ t$

$\text{def NoWithdrawalWhileBlocked}$
 $(C : \text{ComparisonInstance})$
 $(P : \text{Process})$
 $(\tau : \text{Trace } C.\text{RTL_B}) : \text{Prop} :=$
 $\forall t_0 \ t \ \sigma_0 \ \sigma t \ x,$
 $\text{StateAtCycle } \tau \ t_0 \ \sigma_0 \rightarrow$
 $\text{StateAtCycle } \tau \ t \ \sigma t \rightarrow$
 $t_0 \leq t \rightarrow$
 $x \in \text{Front } C \ P \ \sigma_0 \rightarrow$
 $\text{BoundaryReq } C \ x \ \sigma_0 \rightarrow$
 $\neg \text{CommitsBetween } \tau \ x \ t_0 \ t \rightarrow$
 $\text{BoundaryReq } C \ x \ \sigma t \wedge$
 $(\text{IsPushFrontierItem } x \rightarrow$
 $\text{PayloadAtState } C \ x \ \sigma t = \text{PayloadAtState } C \ x \ \sigma_0)$

$\text{def FrontierMinimality}$
 $(C : \text{ComparisonInstance})$
 $(P : \text{Process})$
 $(\tau : \text{Trace } C.\text{RTL_B}) : \text{Prop} :=$
 $\forall t \ \sigma \ x \ y,$

```

StateAtCycle  $\tau \ t \ \sigma \rightarrow$ 
 $x \in \text{Front } C \ P \ \sigma \rightarrow$ 
 $y \in \text{Remain } C \ P \ \sigma \rightarrow$ 
SourceOrderLt  $C \ P \ y \ x \rightarrow$ 
False

```

The next predicates separate two obligations that should not be conflated.

First, there is the $K\varepsilon$ / A11-style WFG-inertness obligation. It applies only to intervals that contain no observable Σ -commit. It says that pure internal / ε -only movement while a process remains blocked at the same frontier cannot create a new wait-for cause.

Second, there is the active drain-only automatic-flush discipline. It permits already-issued downstream work to commit while the process is blocked at a minimal frontier. Such drain commits may be observable Σ -events, so they must not be excluded by an InternalOrEpsilonOnlyBetween premise. The discipline is instead that no new later source work is issued past the blocked frontier, already asserted requests are not withdrawn before commit, and any observable commit owned by the blocked process during the blocked interval is attributable to an already-issued drainable downstream request.

```

def WFGInertForProcess
(C : ComparisonInstance)
(P : Process)
( $\sigma_0 \ \sigma t$  : State) : Prop :=
 $\forall Q,$ 
WFGEdge  $C \ .\text{post } \sigma_0 \ P \ Q \leftrightarrow \text{WFGEdge } C \ .\text{post } \sigma t \ P \ Q$ 

```

```

def NoNewBlockingFrontierCause
(C : ComparisonInstance)
(P : Process)
( $\sigma_0 \ \sigma t$  : State) : Prop :=
 $\forall x,$ 
 $x \in \text{Front } C \ .\text{post } P \ \sigma t \rightarrow$ 
BlockingMessageFrontierItem  $x \rightarrow$ 
ChannelDisabledSide  $C \ .\text{post } x \ \sigma t \rightarrow$ 
 $\exists y,$ 

```

$y \in \text{Front } C . \text{post } P \sigma_0 \wedge$
 $\text{BlockingMessageFrontierItem } y \wedge$
 $\text{SameBlockingCause } C P y x$

$\text{def WFGInertUnderInternalOnlyBetween}$
 $(C : \text{ComparisonInstance})$
 $(P : \text{Process})$
 $(\tau : \text{Trace } C.\text{RTL_B}) : \text{Prop} :=$
 $\forall t_0 t \sigma_0 \sigma t,$
 $\text{StateAtCycle } \tau t_0 \sigma_0 \rightarrow$
 $\text{StateAtCycle } \tau t \sigma t \rightarrow$
 $t_0 \leq t \rightarrow$
 $\text{FrontierBlocked } C P \sigma_0 \rightarrow$
 $\text{FrontierRemainsBlockedOver } C P \tau t_0 t \rightarrow$
 $\text{InternalOrEpsilonOnlyBetween } \tau t_0 t \rightarrow$
 $\text{WFGInertForProcess } C P \sigma_0 \sigma t \wedge$
 $\text{NoNewBlockingFrontierCause } C P \sigma_0 \sigma t$

$\text{def DraggableDownstreamRequest}$
 $(C : \text{ComparisonInstance})$
 $(P : \text{Process})$
 $(\sigma_0 : \text{State})$
 $(x : \text{FrontierItem}) : \text{Prop} :=$
 $\text{AlreadyIssuedBeforeBlockedCutpoint } C P \sigma_0 x \wedge$
 $\text{DownstreamOfBlockedFrontier } C P \sigma_0 x \wedge$
 $\text{BlockingMessageFrontierItem } x \wedge$
 $\text{BoundaryReq } C x \sigma_0$

$\text{def DrainOnlyCommitDiscipline}$
 $(C : \text{ComparisonInstance})$
 $(P : \text{Process})$
 $(\tau : \text{Trace } C.\text{RTL_B}) : \text{Prop} :=$
 $\forall t_0 t \sigma_0 e,$
 $\text{StateAtCycle } \tau t_0 \sigma_0 \rightarrow$
 $t_0 \leq t \rightarrow$
 $\text{FrontierBlocked } C P \sigma_0 \rightarrow$
 $\text{FrontierRemainsBlockedOver } C P \tau t_0 t \rightarrow$
 $e \in \text{SigmaEventsAtCycle } \tau t \rightarrow$

EventOwnedByProcess $C\ P\ e \rightarrow$
 ObservableCommitEvent $e \rightarrow$
 $\exists x,$
 DrainableDownstreamRequest $C\ P\ \sigma_0\ x \wedge$
 EventCommitsFrontierItem $C\ x\ e \wedge$
 $\neg \text{CommitsBeforeCycle } \tau\ x\ t_0 \wedge$
 CommitsAtCycle $\tau\ x\ t$

def DrainOnlyDownstreamProgress
 (C : ComparisonInstance)
 (P : Process)
 ($\tau : \text{Trace } C.\text{RTL_B}$) : Prop :=
 WFGInertUnderInternalOnlyBetween $C\ P\ \tau \wedge$
 DrainOnlyCommitDiscipline $C\ P\ \tau$

structure AutomaticFlush
 (C : ComparisonInstance)
 (P : Process)
 ($\tau : \text{Trace } C.\text{RTL_B}$) where
 block_point :
 $\forall t\ \sigma,$
 StateAtCycle $\tau\ t\ \sigma \rightarrow$
 FrontierBlocked $C\ P\ \sigma \rightarrow$
 FlushBlockedPoint $C\ P\ \tau\ t\ \sigma$

no_new_later_work :
 NoNewLaterWorkWhileBlocked $C\ P\ \tau$

no_withdrawal :
 NoWithdrawalWhileBlocked $C\ P\ \tau$

frontier_minimality :
 FrontierMinimality $C\ P\ \tau$

drain_only_downstream_progress :
 DrainOnlyDownstreamProgress $C\ P\ \tau$

sync_boundary_policy :
SyncBoundaryPolicy C P τ

The sync-boundary policy must explicitly split the ordinary and elaborated cases.
In the elaborated case, it must also state the content of the withholding rule:
producer-facing availability for capacity belonging only to suff_S(s) is unavailable
while the synchronization call s is pending.

def SyncPendingAt
(C : ComparisonInstance)
(P : Process)
(τ : Trace C.RTL_B)
(s : SyncCall)
(t : Cycle) : Prop :=
SyncIssuedButNotCommitted C P τ s t

def FutureEpochCapacityFor
(C : ComparisonInstance)
(P : Process)
(s : SyncCall)
(gateChannel : Channel)
(q : DynMsgInstance) : Prop :=
q $\in$ suffS_msg C P s $\wedge$
MessageOperationInstance q $\wedge$
TransferWouldUseCapacityBehindGate C P s gateChannel q

def SyncBoundaryGateCorrect
(C : ComparisonInstance)
(P : Process)
(τ : Trace C.RTL_B)
(E : ElaborationPackage C.refChoice.outerSysB C.refChoice.Chan_outer C.Sigma_DUT)
(gateChannel : Channel)
(s : SyncCall) : Prop :=
GateChannelOf E gateChannel $\wedge$
 $\forall t \sigma q,$
StateAtCycle $\tau t \sigma \rightarrow$
SyncPendingAt C P τ s t $\rightarrow$
FutureEpochCapacityFor C P s gateChannel q $\rightarrow$

$\text{BoundaryReqPersistentDomain } C \ q \ \sigma \rightarrow$
 $\text{BoundaryReq } C \ q \ \sigma \rightarrow$
 $\text{ChannelDisabled } C \ q \ \sigma$

Here $\text{suffS_msg } C \ P \ s$ is the message-operation-instance projection of $\text{suffS}(s)$.
 The sync-boundary gate rule is stated over DynMsgInstance values because
 BoundaryReq , ChannelEnabled , and ChannelDisabled are obligations of attributed
 message-operation instances, not of arbitrary FrontierItem records.

inductive $\text{SyncBoundaryPolicy}$
 ($C : \text{ComparisonInstance}$)
 ($P : \text{Process}$)
 ($\tau : \text{Trace } C.\text{RTL_B}$)
 | $\text{no_withholding_in_ordinary_SysB} :$
 $C.\text{refChoice.kind} = \text{ReferenceKind.ordinarySysB} \rightarrow$
 $\text{NoProofRelevantSyncBoundaryWithholding } C \ P \ \tau \rightarrow$
 $\text{SyncBoundaryPolicy } C \ P \ \tau$
 | $\text{explicit_gate_in_elab} :$
 $\forall E \ \text{gateChannel } s,$
 $C.\text{refChoice.kind} = \text{ReferenceKind.elaborated} \rightarrow$
 $\text{SyncBoundaryGateCorrect } C \ P \ \tau \ E \ \text{gateChannel } s \rightarrow$
 $\text{SyncBoundaryPolicy } C \ P \ \tau$

In the ordinary Sys_B case, the sync-boundary clause does not add hidden gating.
 It asserts that no proof-relevant sync-boundary withholding is present. If such
 withholding is needed, the theorem instance must use $\text{Sys_B}^{\text{elab}}(E)$, where the
 withholding appears as ordinary ChannelDisabled behavior at an explicit
 sync-boundary / epoch-gate abstract channel.

14. Witness selector and Appendix L

The uploaded document's witness selector is a partial strategy over ε -quiescent source
 cutpoints and ε -modeled choices whose outcomes may affect later Σ -visible behavior; it
 must be causally available from the matched RTL prefix and global Σ -causal history.

Avoid circularity by defining it over the partial construction so far:

structure $\text{WitnessSelector } (C : \text{ComparisonInstance})$ where
 choose :

$(\tau_post_prefix : Prefix\ C.RTL_B) \rightarrow$
 $(\tau_ref_so_far : Prefix\ C.Sys_ref) \rightarrow$
 $(cp : ChoicePoint\ \tau_ref_so_far) \rightarrow$
 Option ChoiceOutcome
 causal :
 $\forall\ \tau_p\ \tau_r\ cp\ out,$
 choose $\tau_p\ \tau_r\ cp = some\ out \rightarrow$
 DependsOnlyOnAvailablePrefix $\tau_p\ \tau_r\ cp\ out$
 consistent_with_post_prefix :
 $\forall\ \tau_p\ \tau_r\ cp\ out,$
 choose $\tau_p\ \tau_r\ cp = some\ out \rightarrow$
 ConsistentWithRTLPrefix $\tau_p\ \tau_r\ cp\ out$

Non-blocking witness mapping.

Failed PushNB / PopNB polls are executed dynamic instances in Q_exec but produce no Σ -event. Therefore the Post $\rightarrow$ Ref construction cannot reconstruct them from the Σ -trace alone. The witness bundle must carry an explicit partial dynamic-instance correspondence for such ε -modeled NB instances.

structure NBInstanceMap (C : ComparisonInstance) where

$\mu_Q :$
 $DynMsgInstance \rightarrow Option\ DynMsgInstance$

maps_only_executed_nb :
 $\forall\ q_post\ q_ref,$
 $\mu_Q\ q_post = some\ q_ref \rightarrow$
 ExecutedInPost C $q_post \wedge$
 ExecutedInRef C $q_ref \wedge$
 $IsNB\ q_post \wedge IsNB\ q_ref$

preserves_owner_and_interface :

$\forall\ q_post\ q_ref,$
 $\mu_Q\ q_post = some\ q_ref \rightarrow$
 $q_post.proc = q_ref.proc \wedge$
 $q_post.channel = q_ref.channel \wedge$
 $q_post.kind = q_ref.kind$

preserves_nb_outcome :

$\forall q_post\ q_ref,$
 $\mu_Q\ q_post = \text{some } q_ref \rightarrow$
 $\text{NBOutcomePost } C\ q_post = \text{NBOutcomeRef } C\ q_ref$

$\text{covers_control_relevant_failed_polls} :$
 $\forall q_post,$
 $\text{ExecutedInPost } C\ q_post \rightarrow$
 $\text{IsFailedNB } q_post \rightarrow$
 $\text{AffectsLaterSigmaOrFrontier } C\ q_post \rightarrow$
 $\exists q_ref, \mu_Q\ q_post = \text{some } q_ref$

$\text{source_order_position_preserved} :$
 $\forall q_post\ q_ref,$
 $\mu_Q\ q_post = \text{some } q_ref \rightarrow$
 $\text{SourceOrderKeyOf } C\ q_post = \text{SourceOrderKeyOf } C\ q_ref$

$\text{respects_source_order_slice} :$
 $\forall P\ q_{1_post}\ q_{2_post}\ q_{1_ref}\ q_{2_ref},$
 $\mu_Q\ q_{1_post} = \text{some } q_{1_ref} \rightarrow$
 $\mu_Q\ q_{2_post} = \text{some } q_{2_ref} \rightarrow$
 $\text{SourceOrderLe } C\ P\ (\text{SourceOrderKeyOf } C\ q_{1_post})\ (\text{SourceOrderKeyOf } C\ q_{2_post}) \leftrightarrow$
 $\text{SourceOrderLe } C\ P\ (\text{SourceOrderKeyOf } C\ q_{1_ref})\ (\text{SourceOrderKeyOf } C\ q_{2_ref})$

$\text{structure WitnessBundle } (C : \text{ComparisonInstance}) \text{ where}$
 $\text{selector} : \text{WitnessSelector } C$
 $\text{nbMap} : \text{NBInstanceMap } C$

$\text{Witness compatibility:}$
 def WC
 $(C : \text{ComparisonInstance})$
 $(W : \text{WitnessBundle } C) : \text{Prop} :=$
 $(\forall \text{ constructionPrefix},$
 $\text{WitnessChoicesRespectMatchedPrefix } C\ W.\text{selector } \text{constructionPrefix}) \wedge$
 $\text{WitnessNBMapRespectsMatchedPrefix } C\ W.\text{nbMap} \wedge$
 $\text{WitnessChoicesAndNBMapAgree } C\ W.\text{selector } W.\text{nbMap}$

Thus WC is the Lean-side packaging of both witness selection and the μ_Q matching obligation. In particular, when E3/E5 reasoning or the $\text{Post} \rightarrow \text{Ref}$

construction refers to failed ε -modeled non-blocking call instances in Q_{exec}, P , the required μ_Q correspondence is supplied by $W.\text{nbMap}$ as part of WC , not as a separate unrelated assumption.

Lemma L.2:

theorem L2_snooping_establishes_WC

(C : ComparisonInstance)

(W : WitnessBundle C)

(hSnoop : SnoopingWrapperAssumptions C W) :

WC C W

`IdentityOrArbitraryNBWitnessBundle C` is the witness bundle used in the NB-CF case. Its selector uses identity choices for ε -modeled non-blocking outcomes that are fixed by the matched prefix, and arbitrary choices for non-blocking outcomes that NB-CF proves cannot affect later Σ -visible control flow or `NextFront` behavior. Its `nbMap` maps matched executed NB instances to the corresponding same dynamic source-call instance when such a match is relevant, and is arbitrary on failed NB instances that NB-CF proves irrelevant.

```
def IdentityOrArbitraryNBWitnessBundle
```

```
(C : ComparisonInstance) : WitnessBundle C :=
```

```
{
```

```
  selector := IdentityOrArbitraryNBSelector C,
```

```
  nbMap := IdentityOrArbitraryNBMap C
```

```
}
```

Corollary L.3:

theorem L3_nb_cf_identity_witness

(C : ComparisonInstance)

(P : Process)

(hNBCF : NBCF C P) :

WC C (IdentityOrArbitraryNBWitnessBundle C)

15. Concrete ObsSigma and deterministic replay

Do not define `ObsSigma` as an informal least fixed point. Define it as a concrete record of the source-state slices required for replay.

```

structure ObsSigma
(C : ComparisonInstance)
(W : WitnessBundle C)
(P : Process)
( $\sigma$  : State) where
control_pos : ControlPos P
relevant_vars : RelevantVarSlice C P  $\sigma$ 
selected_eps_outcomes : SelectedOutcomeSlice C W.selector P  $\sigma$ 
nb_witness_slice : NBWitnessSlice C W.nbMap P  $\sigma$ 
msg_attribution_slice : MsgAttributionSlice C W P  $\sigma$ 
pending_slice : PendingSlice C P  $\sigma$ 
front_slice : FrontSlice C P  $\sigma$ 
nextfront_slice : NextFrontSlice C P  $\sigma$ 
boundary_req_slice : BoundaryReqSlice C P  $\sigma$ 
channel_fact_slice : ChannelFactSlice C P  $\sigma$ 
signal_anchor_slice : SignalAnchorSlice C P  $\sigma$ 
array_mapping_slice : ArrayMappingSlice C P  $\sigma$ 
direct_input_slice : DirectInputSlice C P  $\sigma$ 

```

Here `msg_attribution_slice` records the $Q_{\text{exec}}, P / Q_{\Omega}, P$ attribution facts needed for E3/E5, Issue/Commit replay, WFG construction, and BoundaryReq ownership. It is broader than `nb_witness_slice`: `nb_witness_slice` records μ_Q matching for ε -modeled failed NB calls, while `msg_attribution_slice` records the source-level operation attribution of pending and frontier message-operation instances generally.

Then:

```

def SigmaRelevantEq C W P  $\sigma_1 \sigma_2$  :=
ObsSigma C W P  $\sigma_1$  = ObsSigma C W P  $\sigma_2$ 

```

Replay theorem:

theorem IDR_deterministic_replay

```

(C : ComparisonInstance)
(W : WitnessBundle C)
(hB : BRules C)
(hR : RRules C)
(hE : ERules C)
(hIC : IssueCommitWF C W)

```

(hContracts : ModelingContracts C)
 (hWC : WC C W)
 (P : Process)
 ($\sigma_1 \sigma_2$: State)
 (hObs : SigmaRelevantEq C W P $\sigma_1 \sigma_2$) :
 SameNextSigmaRelevantBehavior C W P $\sigma_1 \sigma_2$

This same ObsSigma object should be reused by Lemma S1 and Corollary J.1.

16. Appendix I per-process construction

Bundle the assumptions:

structure IConstructionAssumptions
 (C : ComparisonInstance)
 (W : WitnessBundle C) where
 wellformed : C.WellFormed
 bRules : BRules C
 rRules : RRules C
 eRules : ERules C
 issueCommitWF : IssueCommitWF C W
 implementation : ImplementationAssumptions C
 witnessCompatible : WC C W
 contracts : ModelingContracts C

Main theorems:

theorem I1_bounded_epsilon_closure
 (C : ComparisonInstance)
 (hB4 : C.assumptions.B.B4_finiteEpsilonDepth)
 (hB5 : C.assumptions.B.B5_quiescentNormalization) :
 ...

theorem I2_channel_progress_instance
 (C : ComparisonInstance)
 (hB3 : C.assumptions.B.B3_channelProgress) :
 ...

theorem I3_finite_extension_step

(C : ComparisonInstance)
 (W : WitnessBundle C)
 (hl : IConstructionAssumptions C W) :
 ...

theorem I4_finite_prefix_construction
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hl : IConstructionAssumptions C W) :
 ...

theorem I5_infinite_limit_construction
 (C : ComparisonInstance)
 (W : WitnessBundle C)
 (hl : IConstructionAssumptions C W) :
 ...

I3 should use IssueAt plus derived IssueTime, not an unconstrained guessed issue cycle.

17. Appendix J system-level correspondence

Define execution scopes:

def Exec_post (C : ComparisonInstance) (τ : Prefix C.RTL_B) : Prop :=
 ∃ τ[∞],
 Extends τ[∞] τ ∧
 InfinitePostExecution C τ[∞] ∧
 FairProgressSatisfying C τ[∞]

def Exec_ref (C : ComparisonInstance) (τ : Prefix C.Sys_ref) : Prop :=
 ∃ τ[∞],
 Extends τ[∞] τ ∧
 RefAdmissible C τ[∞]

Here FairProgressSatisfying includes the B2/B3 fairness/progress assumptions on infinite executions. Deadlock cutpoints are admitted into Exec_post by the B6 idle-stutter convention: once a reachable ε -quiescent deadlock cutpoint is

reached, it may be extended by infinite idle stuttering so that the prefix has an infinite extension. The theorem `deadlock_cutpoint_in_Exec_post` below records that connection explicitly.

Directed compatibility:

Do not define `CompatibleP` or `CompatibleSys` as equality of bucketed executions.

First define the observable units extracted from a bucketed execution. An observable unit may be:

- a single ordinary Σ -event;
- an explicitly atomic multi-endpoint unit, such as a rendezvous Push/Pop pair or a `SyncChannel/ac_sync` pair;
- a same-cycle finite set of independent Σ -events equipped with the partial order actually mandated by E1–E5.

structure `ObservableUnit` (Σ : Type) where

`events` : Finset Σ

`order` : PartialOrderOn events

`atomic` : Prop

`ObservableUnitsOf` extracts Σ -visible events from the bucketed execution and groups them into comparison units as follows:

- an ordinary committed message, signal, or synchronization event may form a singleton observable unit;
- explicitly atomic multi-endpoint events, such as rendezvous Push/Pop pairs and `SyncChannel/ac_sync` handshakes, form one atomic observable unit and may not be split by `CompatibleP` or `CompatibleSys`;
- R2C combinational observations are taken from the ε -quiescent fixed-point bucket and grouped according to the R2C well-formedness relation;
- same-cycle independent events may remain unordered within a unit only when no E1–E5, channel, synchronization, or witness-attribution rule orders them;
- same-cycle events that are merely coincident in one implementation are not forced to remain same-cycle in the other implementation unless they are explicitly atomic or otherwise ordered.

def `ObservableUnitsOf`

(`C` : ComparisonInstance)

(`side` : Side)

```

( $\tau$  : Prefix (SystemOfSide C side)) :
Set (ObservableUnit (C.Sigma)) :=
ExtractSigmaUnits
C side  $\tau$ 
(AtomicRendezvousUnits C side  $\tau$ )
(AtomicSyncUnits C side  $\tau$ )
(R2CFixedPointUnits C side  $\tau$ )
(IndependentSameCycleUnits C side  $\tau$ )

```

HappensBefore is the transitive closure of the order generators that the document requires CompatibleP / CompatibleSys to preserve:

- per-process E1 synchronization order;
- E2 signal-anchor order;
- E3 same-interface order and distinct-interface no-reverse order;
- E4 per-channel FIFO/rendezvous order and capacity-derived dependency order;
- E5 synchronization-boundary preservation;
- explicit atomic-unit membership constraints;
- same-cycle partial order recorded inside a CycleBucket when it reflects a real required dependency rather than an arbitrary linearization;
- witness-selected NB/control dependencies that affect later Σ -visible behavior, NextFront, or boundary-request attribution;
- elaboration-projection constraints when Sys_ref = Sys_B^{elab}(E).

```

def HappensBefore
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=
Relation.TransGen
(HappensBeforeGenerator C side  $\tau$ )
u v

```

```

def HappensBeforeGenerator
(C : ComparisonInstance)
(side : Side)
( $\tau$  : Prefix (SystemOfSide C side))
(u v : ObservableUnit (C.Sigma)) : Prop :=

```

E1SyncOrder C side $\tau \ u \ v \ V$
 E2SignalAnchorOrder C side $\tau \ u \ v \ V$
 E3MessageOrder C side $\tau \ u \ v \ V$
 E4ChannelOrder C side $\tau \ u \ v \ V$
 E5SyncBoundaryOrder C side $\tau \ u \ v \ V$
 AtomicUnitOrder C side $\tau \ u \ v \ V$
 RequiredSameCycleBucketOrder C side $\tau \ u \ v \ V$
 WitnessControlOrder C side $\tau \ u \ v \ V$
 ElaborationProjectionOrder C side $\tau \ u \ v$

CompatibleP : ComparisonInstance $\rightarrow$ Process $\rightarrow$ Prefix C.Sys_ref $\rightarrow$ Prefix C.RTL_B $\rightarrow$ Prop

CompatibleP C P τ_{ref} τ_{post} means:

- the P-local observable units of τ_{ref} and τ_{post} are related by a value-label-preserving bijection;
- explicitly atomic units are mapped to explicitly atomic units and are not split;
- the mapping preserves the P-local happens-before constraints induced by E1, E2, E3, E5, R2/R2C, and the selected witness/request-attribution facts;
- it does not require independent P-local units to occupy the same cycle bucket in both executions.

CompatibleSys : ComparisonInstance $\rightarrow$ Prefix C.Sys_ref $\rightarrow$ Prefix C.RTL_B $\rightarrow$ Prop

CompatibleSys C τ_{ref} τ_{post} means:

- $\forall P, \text{CompatibleP } C \ P \ \tau_{\text{ref}} \ \tau_{\text{post}}$;
- all explicit abstract channels in the chosen Sys_ref / RTL_B comparison satisfy the applicable E4 channel semantics at the agreed abstract boundaries;
- the system-level observable-unit mapping preserves the mandated global happens-before / $\leq_J$ constraints and value labels;
- explicitly atomic multi-endpoint units remain atomic on both sides;
- no equality of CycleBucket lists, no equality of bucket cycle numbers, and no bucket-by-bucket equality of sigmaEvents is required.

Thus the bijection used by CompatibleSys is over ObservableUnitsOf, and its ordering obligation is preservation of HappensBefore. Independent observable units may be placed in different cycle buckets or grouped differently in Sys_ref

and RTL_B , provided their Σ labels, values, atomicity requirements, and required happens-before constraints are preserved.

The notation $\tau_{\text{ref},\text{system}} \approx_{\text{sys}} \tau_{\text{post},\text{system}}$ is an abbreviation for $\text{CompatibleSys } C \ \tau_{\text{ref}} \ \tau_{\text{post}}$. It is a directed compatibility predicate, not a symmetric equivalence relation. The main construction direction is $\text{Post} \rightarrow \text{Ref}$; the reverse direction is restricted to RTL-consistent source prefixes.

J.0:

theorem J0_pairwise_composition

(C : ComparisonInstance)

(W : WitnessBundle C)

(hC : C.WellFormed)

(hB : BRules C)

(hR : RRules C)

(hE : ERules C)

(hIC : IssueCommitWF C W)

(hContracts : ModelingContracts C)

(τ_{ref} : Prefix C.Sys_ref)

(τ_{post} : Prefix C.RTL_B)

(hProc : $\forall P, \text{CompatibleP } C \ P \ \tau_{\text{ref}} \ \tau_{\text{post}}$)

(hChan : ChannelsWellFormedAndMatched C $\tau_{\text{ref}} \ \tau_{\text{post}}$) :
CompatibleSys C $\tau_{\text{ref}} \ \tau_{\text{post}}$

J.1:

structure JAssumptions

(C : ComparisonInstance)

(W : WitnessBundle C) where

wellformed : C.WellFormed

bRules : BRules C

rRules : RRules C

eRules : ERules C

issueCommitWF : IssueCommitWF C W

implementation : ImplementationAssumptions C

witnessCompatible : WC C W

contracts : ModelingContracts C

environment : ReactiveEnvironmentWF C.env

theorem J1_post_to_ref

(C : ComparisonInstance)

(W : WitnessBundle C)

(hJ : JAssumptions C W)

(τ_{post} : Prefix C.RTL_B)

(hExec : Exec_post C τ_{post}) :

$\exists \tau_{\text{ref}}$,

Exec_ref C τ_{ref} $\wedge$

CompatibleSys C τ_{ref} τ_{post}

Restricted reverse:

theorem J1_ref_to_post_restricted

(C : ComparisonInstance)

(W : WitnessBundle C)

(hJ : JAssumptions C W)

(τ_{ref} : Prefix C.Sys_ref)

(hExec : Exec_ref C τ_{ref})

(hConsistent : RTLConsistentSourcePrefix C W τ_{ref}) :

$\exists \tau_{\text{post}}$,

Exec_post C τ_{post} $\wedge$

CompatibleSys C τ_{ref} τ_{post}

Cutpoint correspondence:

structure CutpointCorrespondence

(C : ComparisonInstance)

(σ_{ref} : State)

(σ_{post} : State) where

nextfront_agreement : Prop

frontier_guard_value_boundaryreq_agreement : Prop

buffered_channel_state_agreement : Prop

raw_rendezvous_endpoint_request_agreement : Prop

pending_blocking_enablement_agreement : Prop

```

direct_input_anchor_agreement : Prop
array_mapping_agreement : Prop
r2c_fixed_point_agreement : Prop

theorem J1_cutpoint_correspondence
(C : ComparisonInstance)
(W : WitnessBundle C)
(hJ : JAssumptions C W)
( $\tau_{\text{ref}}$  : Prefix C.Sys_ref)
( $\tau_{\text{post}}$  : Prefix C.RTL_B)
(hCompat : CompatibleSys C  $\tau_{\text{ref}}$   $\tau_{\text{post}}$ )
(hNorm : EpsilonNormalizedPair C  $\tau_{\text{ref}}$   $\tau_{\text{post}}$ ) :
 $\exists \sigma_{\text{ref}} \sigma_{\text{post}}$ ,
TerminalCutpoint  $\tau_{\text{ref}}$   $\sigma_{\text{ref}}$   $\wedge$ 
TerminalCutpoint  $\tau_{\text{post}}$   $\sigma_{\text{post}}$   $\wedge$ 
CutpointCorrespondence C  $\sigma_{\text{ref}}$   $\sigma_{\text{post}}$ 

```

18. Appendix K WFG and liveness

Define WFG over selected Sys_ref boundaries:

```

def WFG
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State) : Digraph Process :=
{ edge := WFGEdge C side  $\sigma$  }

def WFGEdge
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State)
(P Q : Process) : Prop :=
 $\exists q$ ,
q  $\in$  Front C side P  $\sigma$   $\wedge$ 
BlockingMessageFrontierItem q  $\wedge$ 
BoundaryReqSide C side q  $\sigma$   $\wedge$ 
ChannelDisabledSide C side q  $\sigma$   $\wedge$ 
ComplementOwner C side q = Q  $\wedge$ 
InternalChannel C side q.channel

```

Edges are generated from $\text{Front_P}(\sigma)$, not arbitrary non-frontier asserted requests. This intentionally makes the WFG sparser than a graph over all asserted boundary requests: only pending blocking frontier items generate wait-for edges. Failed non-blocking polls and already-asserted non-frontier requests may affect state or occupancy through other rules, but they do not themselves create WFG dependency edges.

```
def ProcessStalledAtFront
(C : ComparisonInstance)
(side : Side)
(P : Process)
( $\sigma$  : State) : Prop :=
Front C side P  $\sigma \neq \emptyset \wedge$ 
 $\forall x,$ 
 $x \in \text{Front C side P } \sigma \rightarrow$ 
BlockingMessageFrontierItem  $x \rightarrow$ 
BoundaryReqSide C side  $x \sigma \rightarrow$ 
ChannelDisabledSide C side  $x \sigma$ 
```

```
def WFGClosed
(C : ComparisonInstance)
(side : Side)
(D : Finset Process)
( $\sigma$  : State) : Prop :=
 $\forall P Q,$ 
 $P \in D \rightarrow$ 
WFGEdge C side  $\sigma P Q \rightarrow$ 
 $Q \in D$ 
```

```
def WFGTerminalSCC
(C : ComparisonInstance)
(side : Side)
(D : Finset Process)
( $\sigma$  : State) : Prop :=
D.Nonempty  $\wedge$ 
StronglyConnectedSubgraph (WFG C side  $\sigma$ ) D  $\wedge$ 
WFGClosed C side D  $\sigma$ 
```

```

def DrainClosed
(C : ComparisonInstance)
(side : Side)
(D : Finset Process)
( $\sigma$  : State) : Prop :=
 $\forall P,$ 
 $P \in D \rightarrow$ 
NoEnabledDrainStepLeavingD C side P D  $\sigma$ 

```

```

def ExistsClosedDrainClosedWFGCycle
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State) : Prop :=
 $\exists D,$ 
WFGTerminalSCC C side D  $\sigma \wedge$ 
DrainClosed C side D  $\sigma \wedge$ 
 $\forall P,$ 
 $P \in D \rightarrow$ 
ProcessStalledAtFront C side P  $\sigma$ 

```

```

def InternalMPDeadlockAt
(C : ComparisonInstance)
(side : Side)
( $\sigma$  : State) : Prop :=
EpsQuiescent  $\sigma \wedge$ 
ExistsClosedDrainClosedWFGCycle C side  $\sigma$ 

```

The K.1 SCC contradiction should use this definition. The closed SCC supplies the cycle of internal message-passing dependencies; DrainClosed rules out escape by downstream draining; ProcessStalledAtFront ensures every process in D is stalled on its current blocking frontier, not merely carrying a non-frontier asserted request.

K assumptions should preserve the document's A1–A11 as a bundle, and add Lean-side auxiliary contracts without pretending they are doc-numbered assumptions.

```

def A11_Kepsilon_WFGInert
(C : ComparisonInstance) : Prop :=

```

$\forall P \tau t_0 t \sigma_0 \sigma t,$
 $\text{StateAtCycle } \tau t_0 \sigma_0 \rightarrow$
 $\text{StateAtCycle } \tau t \sigma t \rightarrow$
 $t_0 \leq t \rightarrow$
 $\text{InternalOrEpsilonOnlyBetween } \tau t_0 t \rightarrow$
 $\text{FrontierBlocked } C P \sigma_0 \rightarrow$
 $\text{FrontierRemainsBlockedOver } C P \tau t_0 t \rightarrow$
 $\text{WFGInertForProcess } C P \sigma_0 \sigma t \wedge$
 $\text{NoNewBlockingFrontierCause } C P \sigma_0 \sigma t$

structure DockAssumptions
 (C : ComparisonInstance)
 (W : WitnessBundle C) where
 A1_point_to_point_channels : Prop
 A2_chosen_Sys_ref_boundaries : Prop
 A3_R_E_rule_conformance : RRules C $\wedge$ ERules C
 A4_B_rules : BRules C
 A5_source_liveness : InternalMPDeadlockFree C.Sys_ref
 A6_issue_commit_wf : IssueCommitWF C W
 A7_witness_bundle : WC C W
 A8_epsilon_normalization : Prop
 A9_single_transfer_per_cycle : Prop
 A10_reset_empty_or_initial_accounting : Prop
 A11_Kepsilon_WFG_inert :
 A11_Kepsilon_WFGInert C

def KElaborationPrerequisite
 (C : ComparisonInstance) : Prop :=
 $\forall c,$
 ChannelCanContributeWFGEdge C c $\rightarrow$
 ChosenAbstractBoundaryOfSysRef C c $\wedge$
 NoHiddenSyncBoundaryGatingAtOrdinaryBufferedBoundary C c $\wedge$
 (RequiresSyncBoundaryWithholding C c $\rightarrow$
 C.refChoice.kind = ReferenceKind.elaborated)

Lean-side additions:

structure KAssumptions
 (C : ComparisonInstance)

(W : WitnessBundle C) where
doc : DocKAssumptions C W
contracts : ModelingContracts C
r2c? : Option (R2C_WF C)
elaboration_prerequisite :
KElaborationPrerequisite C

This avoids fictional A12/A13 labels.

K lemmas:

theorem K0_frontier_blocking_classification
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W)
(σ : State)
(hQ : EpsQuiescent σ) :
...

theorem K0a_rendezvous_endpoint_request_agreement
(C : ComparisonInstance)
(W : WitnessBundle C)
(hCorr : CutpointCorrespondence C σ_{ref} σ_{post})
(c : Channel)
(hB0 : capacity c = 0) :
Req_prod c σ_{ref} = Req_prod c σ_{post} $\wedge$
Req_cons c σ_{ref} = Req_cons c σ_{post}

theorem K1_deadlock_characterization
(C : ComparisonInstance)
(W : WitnessBundle C)
(hK : KAssumptions C W)
(σ : State)
(hReach : Reachable C.RTL_B σ)
(hQ : EpsQuiescent σ) :
InternalMPDeadlockAt C .post $\sigma \leftrightarrow$
ExistsClosedDrainClosedWFGCycle C .post σ

This theorem uses $hK.elaboration_prerequisite$: every channel boundary used by the WFG is the chosen abstract boundary of the Sys_ref / RTL_B comparison after any required elaboration. Synchronization-boundary ramp-down / epoch-gating is therefore represented at explicit Sys_B^{elab} channels, not as a hidden exception to E4 on an otherwise ordinary buffered channel.

theorem K2_dependency_refinement

(C : ComparisonInstance)

(W : WitnessBundle C)

(hK : KAssumptions C W)

($\sigma_ref \sigma_post$: State)

(hCorr : CutpointCorrespondence C $\sigma_ref \sigma_post$) :

WFG C .ref σ_ref = WFG C .post σ_post

theorem K2a_drain_closure_transfer

(C : ComparisonInstance)

(W : WitnessBundle C)

(hK : KAssumptions C W)

(D : Finset Process)

($\sigma_ref \sigma_post$: State)

(hCorr : CutpointCorrespondence C $\sigma_ref \sigma_post$)

(hDrainPost : DrainClosed C .post D σ_post) :

DrainClosed C .ref D σ_ref

Exec-post membership at a deadlock cutpoint should be explicit:

theorem deadlock_cutpoint_in_Exec_post

(C : ComparisonInstance)

(W : WitnessBundle C)

(hK : KAssumptions C W)

(σ_post : State)

(hReach : Reachable C.RTL_B σ_post)

(hDead : ExistsClosedDrainClosedWFGCycle C .post σ_post)

(hIdle : C.assumptions.B.B6_idleStutterTimeDivergence) :

$\exists \tau_post$,

TerminalCutpoint $\tau_post \sigma_post \wedge$

Exec_post C τ_post

Main K theorem:

theorem K1_liveness_preservation

(C : ComparisonInstance)

(W : WitnessBundle C)

(hK : KAssumptions C W) :

InternalMPDeadlockFree C.Sys_ref →

InternalMPDeadlockFree C.RTL_B

This is intentionally one-way. Do not state:

InternalMPDeadlockFree C.Sys_ref ↔ InternalMPDeadlockFree C.RTL_B

19. Appendix R facade

Appendix R should be formalized last as aliases/wrappers over earlier theorems.

Use Lean names that avoid the doc's R.2 collision:

theorem R_lemma1_front_nextfront_classification := ...

theorem R_theorem1_prefix_matching := ...

theorem R_corollary1_cutpoint_correspondence := ...

theorem R_corollary1a_rendezvous_endpoint_request_agreement := ...

theorem R_lemma2_dependency_refinement := K2_dependency_refinement

theorem R_theorem2_closed_cycle_and_one_way_drain_transfer := ...

Do not call cutpoint correspondence R1_cutpoint_correspondence; in the doc structure, it corresponds to a corollary, not Theorem R.1.

20. What remains as hypotheses

These remain theorem parameters or design/tool obligations:

1. R-rules R1, R2, R2C, R3, R4.
2. E-rules E1–E5.
3. B1–B6, with B2/B3 encoded temporally.
4. Implementation assumptions such as I.A0.
5. Witness compatibility WC, derived from Lemma L.2 or Corollary L.3 where applicable.

6. Direct-input contracts.
 7. Array access mapping rules.
 8. Elaboration package E, when $\text{Sys_ref} = \text{Sys_B}^{\text{elab}}(E)$.
 9. Source-level internal message-passing deadlock freedom for K.
-

21. What should be proved inside Lean

Lean should prove:

- bucketed execution infrastructure;
- temporal lemmas for AlwaysFrom / EventuallyFrom;
- channel-capacity lemmas for rendezvous and buffered cases;
- issue existence/uniqueness and derived IssueTime;
- direct-input late-sampling preservation from primitive environment contracts;
- array sync-boundary commit-order preservation;
- R2C fixed-point observation lemmas from R2C_WF;
- automatic-flush frontier lemmas;
- Lemma L.2: snooping establishes WC;
- Corollary L.3: NB-CF identity witness;
- I.DR over concrete ObsSigma;
- Appendix I finite and infinite witness construction;
- Proposition J.0;
- Theorem J.1 post-to-reference;
- restricted reverse theorem for RTL-consistent source prefixes;
- Corollary J.1 cutpoint correspondence;
- K.0/K.0a/K.1/K.2/K.2a;
- deadlock cutpoint membership in Exec_post;
- Theorem K.1 one-way liveness preservation;

- Appendix R aliases.
-

22. Development milestones

Milestone A — Sequential, blocking-only, non-elaborated core

Scope:

- Sys_ref = Sys_B;
- sequential processes only;
- no NB calls;
- no snooping;
- no external arrays;
- no direct-input late sampling;
- no combinational R2C;
- bucketed execution model.

Targets:

J1_post_to_ref_blocking

J1_cutpoint_correspondence_blocking

K1_liveness_preservation_blocking

Milestone B — Temporal B2/B3

Add:

- infinite bucketed executions;
- AlwaysFrom, EventuallyFrom;
- concrete FairAction;
- P_push, P_pop.

Target:

K1_liveness_preservation_with_temporal_progress

Milestone C — Non-blocking under NB-CF

Add:

- Qexec;
- failed NB polls as ϵ ;
- successful NB transfers as Σ ;
- Corollary L.3.

Target:

J1_post_to_ref_nb_cf

Milestone D — Witness selector / snooping

Add:

- partial-prefix WitnessSelector;
- Lemma L.2.

Target:

J1_post_to_ref_with_snooping

Milestone E — Direct inputs

Add:

- DirectInputContract;
- derived lateSamplingPreservesLogicalAnchor;
- bucket placement for logical anchor vs physical late sample.

Target:

J1_post_to_ref_direct_inputs

Milestone F — External arrays

Add:

- MemoryEvent;
- ArrayAccessMapping;
- sync-relative array commit rules.

Target:

J1_post_to_ref_external_arrays

Milestone G — R2C

Add:

- `combNetwork?`;
- `CombNetwork`;
- `R2C_WF`;
- fixed-point bucket observation semantics.

Target:

`J1_post_to_ref_with_R2C`

Milestone H — Elaboration

Add:

- `ElaborationPackage Sys_B Chan_outer  $\Sigma$ _DUT`;
- `$\Pi$ _elab`;
- `$\pi_{\Sigma,E}$` ;
- `A_E`;
- explicit-channel interpretation of `B`, `occ`, `BoundaryReq`, `ChannelEnabled`, `Front`, and `WFG`.

Targets:

`J1_post_to_ref_elab`

`K1_liveness_preservation_elab`

Milestone I — Appendix R facade

Add compact R theorem wrappers after the full proof stack is available.

23. Expected difficulty

The hardest pieces remain:

1. Bucketed execution and ideal induction.
2. Temporal B2/B3 over infinite executions.
3. Concrete `ObsSigma` and `I.DR`.

4. Appendix J finite-ideal construction.
5. Direct-input late-sampling contracts.
6. External array access mapping.
7. R2C fixed-point semantics.
8. Appendix Q elaboration obligations.
9. Appendix K Exec_post membership at deadlock cutpoints.

A polished full formalization is plausibly in the 18k–30k Lean line range if direct inputs, arrays, R2C, temporal fairness/progress, non-blocking witness selection, snooping, automatic flush, Issue/Commit attribution, A11/K ϵ , and elaboration are all included. A core sequential/blocking/non-elaborated proof would be substantially smaller, plausibly 6k–10k lines; an intermediate blocking-plus-temporal-WFG proof would likely fall around 10k–16k lines.